



1. Lee las instrucciones

3. Pincha en el numero para introducir un nuevo caso de uso en la categoría elegida

2. Si alguien ya ha escrito un caso de uso como el tuyo y en la misma categoría, le das al like en lugar de escribirlo de nuevo

¿Qué casos de uso de IA generativa tienes en tu tarea como INVESTIGADOR-A (ej: "para revisar artículos")? Pincha en el numero de la categoría que corresponda y comparte cosas que mantendrías (Keep), cosas que no te han funcionado bien y no volverías a usar (Stop) y cosas que aún no has probado y te gustaría intentar algún día (Start). Contesta tantos casos de uso como quieras.

S

1

Start

K

2

Keep

S

3

Stop

Respuestas Todos

3 Revisar Los artículos que me asignan como Reviewer

2 Revisar mi artículo antes de mandarlo a revista

3 Extracción de contenido en revisiones sistematicas

1 Cribado en revisiones sistematicas

2 Mejora de redaccion de Texto académico



- Mientras esperas, accede a este QR y completa la actividad

<https://app.wooclap.com/IRPZIH/questionnaires/686a5f07148b366293f12188>

Objetivos del taller. Aplicación de IA para la investigación científica



- Comprender los beneficios y limitaciones del uso de IA en la investigación académica.
- Identificar herramientas y técnicas de IA útiles para el análisis de datos, revisiones de literatura y otras tareas propias de investigación.
- Estimular la reflexión crítica sobre el impacto de la IA en la producción y validación del conocimiento.

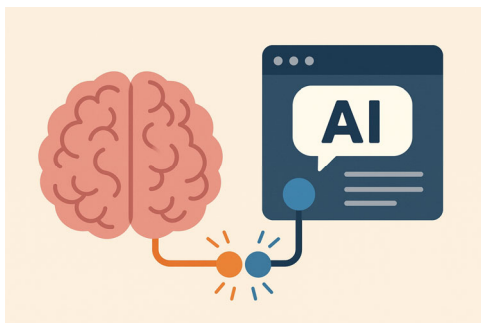
¿Qué voy a hacer yo en la charla? → contar historias, mostrar ejemplos
¿Qué vas a hacer tu? → interrumpirme y preguntarme lo que te inquiete, interese, dudas



Primera parte

- Algunos posibles contenidos (modificables en función de tus intervenciones)

- Concienciación. Algunos mantras [El tiempo que perdiste por tu rosa, hace que tu rosa sea tan importante] et al.
- Recomendaciones para el uso de IA generativa en investigación
- Conceptos básicos son fundamentales para entender y aplicar la IA en la investigación científica



Segunda parte

- ¿Qué? Describir que herramientas IA generativa usamos en investigación
- ¿Para Qué? Describir los principales casos de uso donde las usamos
- Ejemplos de algunos caso de uso donde la IA gen nos ¿ayuda? a realizar investigación

- <https://github.com/jamg-upv/genAI4ResearchTeaching/tree/main/PDGENIsevilla>



Cultura IAgen



Es de RRHH pero puedes extrapolarse a cualquier área de managemente. Los editores de esta revista “ven bien” que se investigue el impacto de IA en la Gestion. “no ven bien” que se investigue con IA.. [Cambiarán de punto de vista antes de 2 años]



El estado actual de las cosas, en un estudio de una temática analizando como un humano ejecuta ideas (de humano o IA). No sabemos cual es el execution cuando lo hace una IA

<https://arxiv.org/abs/2506.20803>

Large Language Models (LLMs) have shown promise in accelerating the scientific research pipeline. A key capability for this process is the ability to generate novel research ideas, and prior studies have found settings in which LLM-generated research ideas were judged as more novel than human-expert ideas. However, a good idea should not simply appear to be novel, it should also result in better research after being executed. To test whether AI-generated ideas lead to better research outcomes, we conduct an execution study by recruiting 43 expert researchers to execute randomly-assigned ideas, either written by experts or generated by an LLM. Each expert spent over 100 hours implementing the idea and wrote a 4-page short paper to document the experiments. All the executed projects are then reviewed blindly by expert NLP researchers. Comparing the review scores of the same ideas before and after execution, the scores of the LLM-generated ideas decrease significantly more than expert-written ideas on all evaluation metrics (novelty, excitement, effectiveness, and overall; $p < 0.05$), closing the gap between LLM and human ideas observed at the ideation stage. When comparing the aggregated review scores from the execution study, we even observe that for many metrics there is a flip in rankings where human ideas score higher than LLM ideas. This ideation-execution gap highlights the limitations of current LLMs in generating truly effective research ideas and the challenge of evaluating research ideas in the absence of execution outcomes.



	Novelty	Excitement	Effectiveness	Overall
Human Condition Ideation Score	4.912	4.404	4.833	4.596
AI Condition Ideation Score	5.778	5.653	6.003	5.382
<i>p</i> -value (FDR)	0.035*	0.004**	0.001**	0.035*
Human Condition Execution Score	4.903	4.482	4.782	3.968
AI Condition Execution Score	4.729	3.896	4.125	3.406
<i>p</i> -value (FDR)	0.586	0.175	0.266	0.175

Table 4: Comparison of mean ideation vs. execution scores for human and AI conditions. We treat the average score of each idea as an independent data point, so the sample size is $N = 19$ for the human condition and $N = 24$ for the AI condition. For the p -values, we perform two-sided t-tests with FDR correction. * means $p < 0.05$, ** means $p < 0.01$.

Recomendaciones para tareas de publicación en Revistas. Esto cambia por meses (cada vez hay mas usos que no hay que declarar). No es para HACER la investigación, son para escribir el PAPER

https://s3.eu-west-2.amazonaws.com/stm.offloadmedia/wp-content/uploads/2025/04/07023544/STM_AI_Classification_Report_April2025.pdf

	Use of AI in the preparation of academic manuscripts (*)	What	Examples	Does NOT include
1	Refine, correct, edit, or format the manuscript to improve clarity of language (**)	Automated tools were used to suggest language improvements	Spell checkers, grammar checkers, predictive text in Microsoft Word	Generating text using AI prompts, generating machine summaries of text, analysing or summarising textual documents as part of the research process
2	Writing or drafting manuscript content	AI was used to create part or all of the manuscript text	Generating text using AI prompts, asking AI to expand on text, generating machine summaries of text (e.g. to summarise arguments made in another publication)	Use of simple spelling and grammar checkers, analysing or summarising textual documents as part of the research process
3	Translation of manuscript text for the purpose of publishing (***)	AI was used to assist translation of an author's original work into a secondary language	Authors using tools (e.g. Google translate, ChatGPT) to assist with translating their manuscript draft	Translation of materials as part of the research process (unrelated to manuscript preparation)
4	Refining or formatting of data reported in the submitted manuscript	Data reported in the submitted manuscript (e.g. included as supplementary material, or stored in a data repository linked to from the manuscript) are refined or formatted using AI	Authors using tools to correct (e.g. on the level of language) or format research data before it's published	Using AI to visualise data sets (activity 6) or data manipulation; any generation, correcting, or editing of data used as part of the research process
5	Generation, refinement, correction, editing or formatting of images, diagrams or other figures for illustrative purposes only	AI was used to generate images or diagrams or other figures in the manuscript for illustrative or aesthetic purposes only	Using AI to create an image of e.g. people farming in an ancient society - the sort of image that might otherwise be created by a human artist as an illustration	Visualisation of actual data or research outputs, editing of visualisations of actual data or results, presentation of generated images as research outputs in themselves or as representing research outputs
6	Generation, refinement, correction, editing or formatting of visualisations of research data or results	AI tools were used to visualise or refine visualisations of research data/results	Using AI to generate a graph or other visualisation of a given research data set	Generation of images or other content without any basis in real research data/outputs
7	Refinement or formatting of code reported in the submitted manuscript.	AI tools were used to assist with refinement of the presentation of code within the manuscript	Using AI to assist with human assessment to improve the readability of code submitted as part of a manuscript	Code generated or improved by AI that was used for the research itself; any new code generated by AI alone
8	Assisting with compilation of reference lists	AI tools were used to create or suggest references to include in the manuscript	Asking AI to generate a list of articles that are related to the research to use as references	AI generation of plausible-sounding references that do not exist; using automated tools to generate a citation in a particular style/format based on a DOI
9	Presentation of any kind of content generated by AI as though it were original research data/results from non-machine sources (****)	AI was used to create data, text, images, graphs, spectra, or other content that is presented as though it were original research data/results collected or analysed from other, non-machine sources	Any content generated wholesale by AI without any basis in original research data/outputs, presented as though it is based on original data/results	Using AI to visualise original research data/results (activity #6); carrying out research on the outputs of GenAI (in which case GenAI outputs can be considered research data)

(*) Note that use of AI in other stages of research processes is outside the scope of this classification. This classification considers only activities that use AI to assist with the preparation of manuscripts.

(**) 2023 STM Guidelines recommend: disclosure not necessary

(***) It is understood that, under copyright law, it remains the right of the right holder to authorize, or not, a work's translation

(****) 2023 STM Guidelines recommend: this use of AI not allowed

Por lo tanto... necesitamos algunas recomendaciones de uso. Esta es una propuesta (el borrador que trabajé para la UPV). En rojo las aplicaciones que incluyo en el webinar



Por ejemplo, este video de 2 minutos fue creado con una Iagen a principios de 2024

<https://youtu.be/GK7qZmfRsl0>

- Es importante que conozcas y comprendas el potencial de las herramientas de Inteligencia Artificial para tu trabajo, pero también que seas consciente de sus limitaciones y de los desafíos que plantea su uso. Te proponemos algunas recomendaciones específicas para la investigación en la UPV:
 - Familiarízate con las capacidades y limitaciones de las herramientas de IA: Entiende cómo pueden ayudarte en tareas como la búsqueda de información, la generación de ideas, la redacción de textos, etc., pero también reconoce sus deficiencias en cuanto a la comprensión profunda, la creatividad y el pensamiento crítico.
 - Integra la IA de manera responsable y equilibrada. No dependas excesivamente de estas herramientas, y úsalas como complemento a tus propias habilidades y conocimientos. Mantén un enfoque crítico decidiendo qué tareas pueden ser automatizadas y cuáles no. Por ejemplo, puedes utilizar la IA para automatizar tareas repetitivas y/o administrativas.
 - Promueve la transparencia y la atribución adecuada citando correctamente el contenido generado directamente por la IA (plataforma, versión y, cuando proceda el prompt utilizado y otros parámetros de interés como, por ejemplo, temperatura), con el fin de mantener la integridad/reproducibilidad de tu investigación y contribuir a la confianza en la comunidad científica.
 - Asegúrate de que el uso de las herramientas de IA que utilices no genere sesgos o discriminación.
 - Aprovecha los recursos y soportes ofrecidos por la universidad y otras instituciones para estar al tanto de las últimas tecnologías y sus aplicaciones en tu campo de estudio. Para ello puedes participar en proyectos piloto y evaluaciones para construir una base de evidencia: contribuye a la realización de estudios que permitan evaluar el impacto y la eficacia del uso de la IA en diferentes contextos de investigación
 - Explora cómo las herramientas de IA pueden facilitar la transferencia del conocimiento generado en la investigación
 - Protege la privacidad y seguridad de los datos cuando uses herramientas de IA, especialmente al manejar datos confidenciales o sensibles. Utiliza plataformas que cumplan con las regulaciones de protección de datos (como la GDPR) y evita compartir información sensible en plataformas públicas.
 - Las herramientas de IA pueden ayudarte a identificar patrones y relaciones en los datos que podrían pasar desapercibidos, estimulando así la innovación y la creación de conocimiento de excelencia.

Algunas Research Tools (no con animo de ser exhaustivo, sino para que se vea que hay más de 70)



<https://claude.ai/public/artifacts/143567e5-3fb8-49a4-bf1a-729fa0da0ec8>

Research Tools Analytics

Dashboard de herramientas bibliográficas basadas en IA

73

TOTAL HERRAMIENTAS

48

NUEVAS AGREGADAS

11

CATEGORIAS

30

HERRAMIENTAS GRATUITAS

Filtros Inteligentes

CATEGORIA

Todas las categorías

MODELO DE PRECIO

Todos los precios

SOPORTE PDF

Cualquiera

REQUIERE REGISTRO

Cualquiera

IDIOMA

Todos los idiomas

BUSQUEDA RAPIDA

Buscar herramienta o función

APLICAR FILTROS

LIMPIAR TODO

TOP RECOMENDADA

SOLO NUEVAS

Mostrando todas las 73 herramientas • 48 nuevas agregadas

DESCARGAR EXCEL

DESCARGAR CSV

GENERAR REPORTE

Base de Datos de Herramientas

Análisis completo de herramientas para investigación académica

CATEGORIA	HERRAMIENTA	FUNCION PRINCIPAL	PRECIO	REGISTRO	PDF	IDIOMA	DESCRIPCION
BUSQUEDA Y SINTESIS DE LITERATURA	Consensus	Buscar y resumir papers	Gratis	Si	No	Inglés	IA que responde preguntas con registros científicos
BUSQUEDA Y SINTESIS DE LITERATURA	Semantic Scholar	Buscar y resumir papers	Gratis	No	No	Inglés	Motor académico con resumen y citas clave
BUSQUEDA Y SINTESIS DE LITERATURA	Scopus AI	Buscar y resumir papers	De pago	Si	No	Inglés	Síntesis de artículos de investigación académica
BUSQUEDA Y SINTESIS DE LITERATURA	SciSpace	Análisis de PDF	Freemium	Si	Si	Inglés	Responde preguntas y explica artículos
BUSQUEDA Y SINTESIS DE LITERATURA	Elicit	Preguntas de investigación	Gratis	No	No	Inglés	Responde con papers relevantes

UPV

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doe
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SINCCE

47

Esta es la comparación que hace “scite” que es “juez y parte” y en la comparación que divulgó en junio estaba comparando modelos de hace 2 o 3 años en algunas plataformas con modelos de hace 8 meses (no es comparable al ritmo que van las cosas)

Comparing Research Specific Tools

scite_ RS Research Solutions

Summary of “Emerging AI Tools for Literature Review: Comparison of GenAI Tools” Hong Kong University of Science & Technology – Updated June 2025

Tool	Functionality	Source Data	Strengths	Limitations
Scite Assistant	RAG with citation-based summaries	Publisher Agreements and full text , OA Fulltext and all Journal Metadata	Fact-checked summaries, search transparency	Not 100% full text yet
Elicit	Summaries, matrices, long-form reports	Semantic Scholar (OA full text, others' abstracts)	Lit synthesis matrix, filtering options	Subscription for some features
SciSpace	Summaries, matrices, Deep Review	OA full text	Filtering, Deep Review mode	Source not detailed
Consensus	Summaries with Consensus Meter	Semantic Scholar	Semantic + keyword hybrid search	Quota for advanced features
Undermind	Iterative search + relevance scoring	Semantic Scholar (titles/abstracts)	High recall, relevance explanations	Limited search quota, hidden records
AI2 Paper Finder	Semantic search + citation trails	Semantic Scholar	Follows citation chains, relevance summaries	Not full summaries per query
AI2 ScholarQA	Long-form reports with quotes	Semantic Scholar (8M full text)	Quote extraction, structured outlines	Abstracts only for most papers
Primo	RAG with Boolean + embedding rerank	Metadata/abstracts	Integrated with Primo, wide coverage	Only 5 results, weak query logic
WOS AI Assistant	RAG + citation network visualizations	Web of Science Core	WOS-based, visual tools, curated content	Extra cost, requires WOS sub
Scopus AI	RAG with topic experts & concept map	Scopus (2003+)	Concept maps, foundational papers	No prefiltering, extra cost
ScienceDirect AI	RAG with chat + full-text summaries	ScienceDirect full-text	Chat with paper, uses full-text	Limited articles per query
Next by	Web search + Deep Research	Open web (Some Wiley now)	Long-form answers, broad coverage	Quota for deep research

Igual después de estos “mantras” decides que no vale la pena seguir escuchándome



- No es lo que la IA pueda hacer, sino lo que me interesa a mi que haga
 - A mi me gusta escribir, no quiero que la IA escriba por mi
 - A mi me gusta pensar, no quiero que la IA piense por mi
 - A mi me gusta tomar decisiones, no quiero que la IA tome decisiones por mi
 - Si no te gusta escribir, pensar y tomar decisiones, igual es que no te gusta investigar

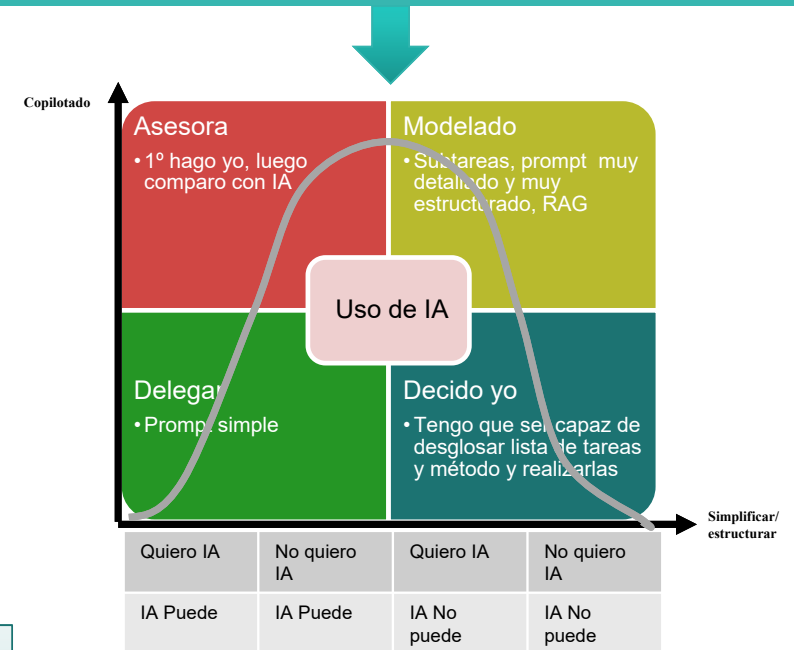
Modelo VAHB para decidir qué hacer y quien lo hace

- Cualquier “proyecto/tarea” podemos partirlo en tareas
- Las NVA evitables deberíamos NO hacerlas ni nosotros ni la IA

Este mapa será ligeramente distinto cada vez porque no siempre coincidiremos en qué es valor añadido y qué hace la IA mejor que “yo”

¿Cómo decido qué delegar?

Lo que es VA, NVAn etc. depende de mi rol y/o de la contribución de mi aprendizaje
→ La combinación adecuada depende de qué tarea, y del valor añadido de hacerla tu misma-o o no

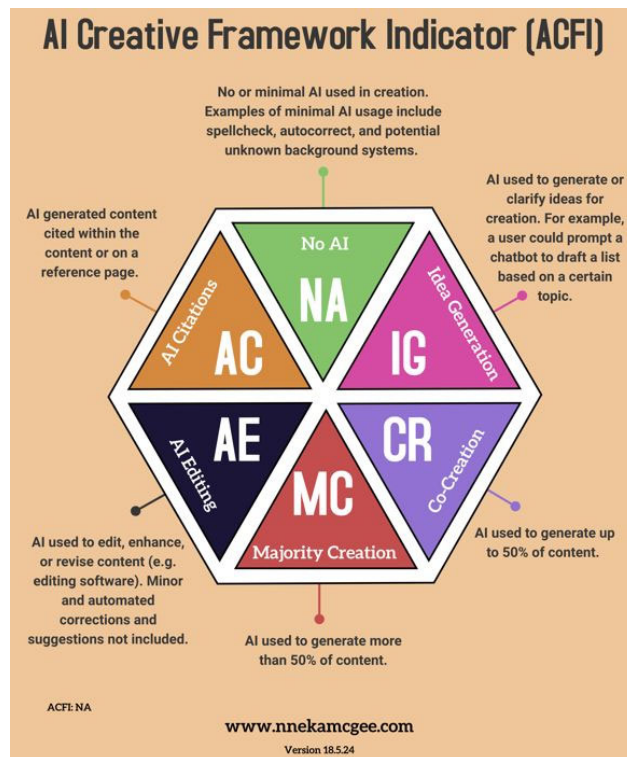


¿CÓMO UTILIZAS LA INTELIGENCIA ARTIFICIAL?

La IA te hace todo el trabajo sin que tengas que pensar	La IA te escribe el contenido y tú lo editas	La IA te ofrece contenidos y tú los mejoras
La IA te crea imágenes y tú eliges las mejores para tu proyecto	Antes ideas básicas y la IA te hace un borrador	Mientras escribas, la IA te da ideas y te guía
Escribes y la IA te ayuda a lo largo del proceso	Escribes un borrador, la IA corrige, y las mejoras	Buscas datos con la IA, para escribir el trabajo solo
Conseñas ideas a la IA, pero haces el trabajo solo	Escribes todo el trabajo y la IA lo revisa y mejora	Haces todo el trabajo sin ayuda de la IA

<https://iaenlasaulas.blogspot.com/2024/05/como-utilizas-la-ia-una-vez-para.html>

Con lo de la diapo anterior elegimos el grado de “copilotado” que queremos con IA



<https://leonfurze.com/2024/05/20/the-ai-assessment-scale-in-action-examples-from-k-12-and-higher-education-across-the-world/>

¿CÓMO UTILIZAS LA INTELIGENCIA ARTIFICIAL?

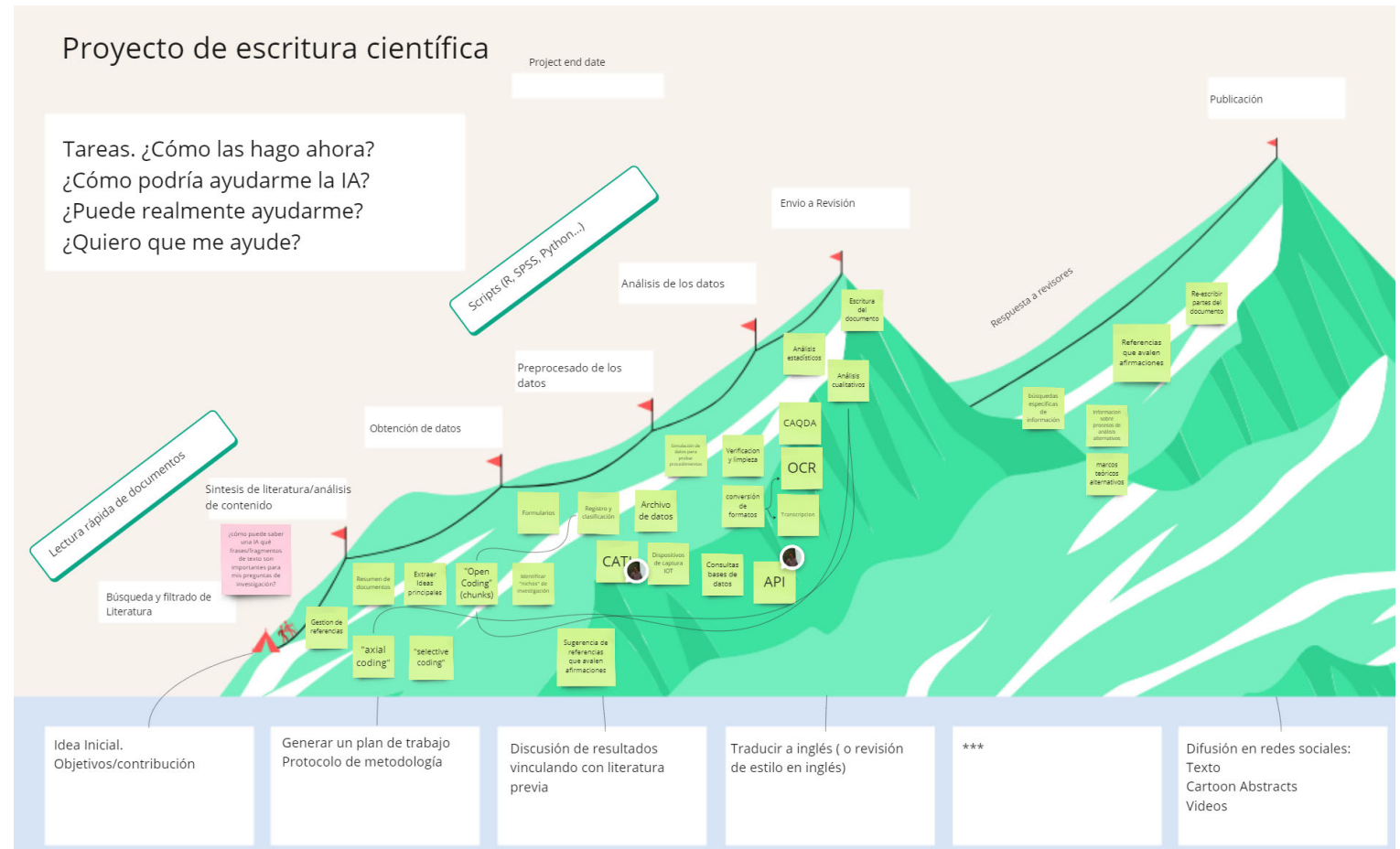
100% 0% La IA te hace todo el trabajo sin que tengas que pensar	95% 5% La IA te escribe el contenido y tú lo editas	90% 10% La IA te ofrece contenidos y tú los mejoras
80% 20% La IA te crea borradores y tú eliges las mejores partes	70% 30% Anotas ideas básicas y la IA te hace un borrador	60% 40% Mientras escribes, la IA te da ideas y te guía
50% 50% Escribes y pides ayuda a la IA si te atascas	40% 60% Escribes un borrador, la IA otro, y los comparas	30% 70% Buscas datos con la IA, pero escribes el trabajo solo
20% 80% Consultas ideas a la IA, pero haces el trabajo solo	10% 90% Escribes todo el trabajo y la IA lo revisa y mejora	0% 100% Haces todo el trabajo sin ayuda de la IA

<https://iaenlasaulas.blogspot.com/2024/05/como-utilizas-la-ia-una-guia-para.html>

Si lo delegas todo en la IA, ¿qué aprendes tú? ¿cual es tu contribución?

Vayamos a las etapas de un proyecto de publicación científica

<https://miro.com/app/board/uXjVka6lG44=?moveToWidget=3458764589004469814&cot=14>



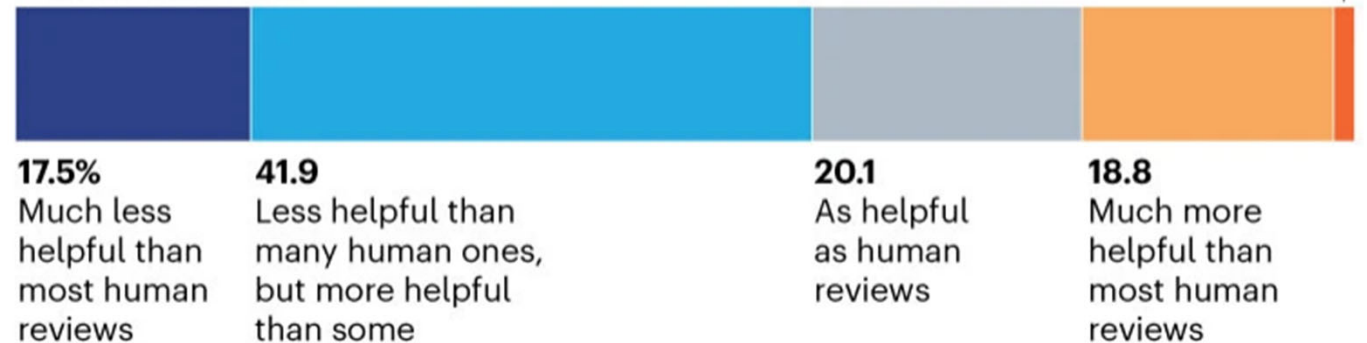
Y si resulta que la IA es tan útil, o casi, que un humano

<https://www.nature.com/articles/d41586-025-00894-7>

COMPARING AI AND HUMAN PEER REVIEW

A survey asked 300 researchers to compare human and LLM* reviews of their own papers. Respondents found LLM feedback slightly less helpful, on average, than human feedback, but LLMs performed better than some human reviewers.

LLM reviews were:



*Large language model; in this case, GPT-4, from the firm OpenAI. Percentages do not add to 100% owing to rounding.

©nature

¿hay alguna de estas palabras qué no sabéis que significan? Os lo dejo como deberes para que lo leais y no gasto el tiempo de la sesión ahora

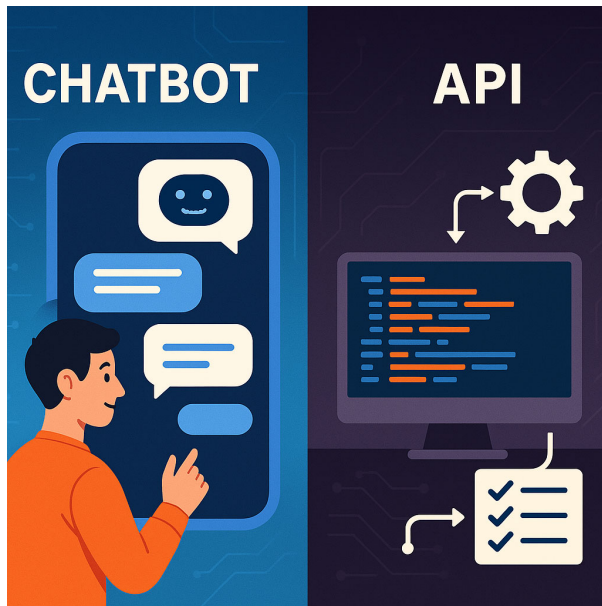
- User prompt
- System prompt
- Temperature
- Context
- RAG
 - Chunk Embeddings
- NLP
 - Sentence Transformers
 - Embeddings
 - Distance (cosine)
 - Generative Pretrained Transformers
 - Attention mechanism

Explicación en lenguaje común que espero que todo el mundo pueda entender

- **User prompt** (Lo que tú me dices)
 - Es como cuando tú haces una pregunta o pides algo. Por ejemplo, "cuéntame un cuento" o "ayúdame con mi tarea". Es lo que TÚ escribes para hablar la IAgén.
- **System prompt** (Las reglas secretas que tengo)
 - Es como las reglas que los programadores dieron a la IAgén antes de que pudiéramos hablar. Por ejemplo, "sé amable", "ayuda siempre", "no digas groserías". Tú no puedes ver estas reglas, pero la IAgén siempre las sigue
 - En algunos casos (proyectos, "chat builder" o uso del LLM por API con un script) puedes "controlar" el System prompt (añadirlo al programado o, en algunos modelos, sustituir el programado)
- **Temperature** (Qué tan creativo soy)
 - Imagínate que la IAgén tenga un botón de creatividad. Si está en "frío", siempre da respuestas muy parecidas y serias. Si está en "caliente", es más divertida, creativa, impredecible, pero a veces digo cosas raras. Es como elegir entre ser muy formal o muy juguetón.
- **Context** (Lo que recordamos de nuestra conversación)
 - Es como nuestra memoria de la conversación. Si le dijiste a la IAgén que te gusta el helado de chocolate, lo recuerda para seguir hablando contigo sobre eso. Es todo lo que hemos dicho antes en nuestra charla (hasta el límite que los programadores hayan establecido)
 - Algunas plataformas (como POE) te permiten indicar tu la amplitud del contexto
- **RAG** (Buscar información extra)
 - Es como cuando no sé algo y voy a buscar en una biblioteca especial para darte mejor información. En lugar de solo usar lo que ya sé, voy a buscar datos frescos para ayudarte mejor.
 - **Chunk Embeddings** (Pedacitos de información organizados)
 - Imagínate que tienes muchos libros y cortas cada página por cada párrafo. Luego, cada párrafo lo conviertes en un vector (una lista de números). Así la IAgén puede encontrar el párrafo que necesito cuando preguntas algo. Por menos distancia con la pregunta
- **Embeddings**
 - Imagínate que quieres describir a tu mejor amigo. Podrías decir:
 - Lo alto es (del 1 al 10)
 - Lo divertido es (del 1 al 10)
 - Lo bueno es en matemáticas (del 1 al 10)
 - Lo deportista es (del 1 al 10)
 - Entonces tu amigo sería algo como: [7, 9, 5, 8] - esos son 4 números que lo describen-.
 - Ahora imagínate que en lugar de 4 cosas, quisieras describir TODAS las características posibles de tu amigo: su humor, inteligencia, creatividad, bondad, si le gustan los animales, si es tímido, si le gusta la música... podrían ser 300 o 1000 características diferentes
 - Eso es exactamente lo que hace un embedding con las palabras. Toma una palabra como "gato" y la convierte en una lista súper larga de números (como [0.2, -0.5, 0.8, 0.1, -0.3...]) donde cada número representa una característica de esa palabra.
 - La palabra "perro" tendría números muy parecidos a "gato" porque ambos son animales peludos y mascotas. Pero "avión" tendría números muy diferentes.
 - Vector n-dimensional
 - Es el nombre técnico para esa lista súper larga de números. Si tiene 300 números, decimos que es un "vector de 300 dimensiones". Es como si cada palabra viviera en un espacio gigante con 300 direcciones diferentes, y el vector representa las coordenadas que nos dicen dónde está exactamente en ese espacio.
 - Por eso las palabras parecidas "viven cerca" en ese espacio invisible, y las diferentes "viven lejos".
- **Distancia de coseno**
 - Una forma de medir la distancia donde lo que importa es la Dirección (no la distancia "euclídea"). Si los vectores apuntan en la misma dirección tienen menos distancia (aunque uno sea más corto o más lejano)

NLP, Sentence Transformers y GPTs

- **NLP** (Entender el lenguaje humano)
 - La capacidad de la IAgén para entender lo que se le dice y responderte en tu idioma. Es como ser un traductor súper inteligente que entiende no solo las palabras, sino también lo que realmente quieres decir.
 - Sentence Transformers vs GPT (Dos tipos diferentes de robots inteligentes)
- **Los Sentence Transformers** son como robots que son súper buenos para entender y comparar frases. Son como bibliotecarios que pueden encontrar el libro que “**CREEN**” que buscas a partir de una información incompleta que les das. Convierten el texto en números (y siempre los mismos números para el mismo texto) en base a los pesos de su entrenamiento. Convierten frases nuevas en embeddings en tiempo real. Cuando les das una frase que nunca han visto antes, la procesan y crean un vector nuevo específicamente para esa frase completa.
 - Su trabajo es crear representaciones numéricas de frases completas
 - Son especialistas en capturar el significado de oraciones enteras
- **Los GPT** son como robots súper buenos para crear y escribir cosas nuevas. Durante el entrenamiento, ya se calcularon y “congelaron” todos los embeddings de los tokens. Cuando tú escribes algo, tus palabras se convierten en tokens, cada token ya tiene su embedding calculado, Los pesos de todas las conexiones también estaban ya calculados. Solo se comparan los embeddings para seleccionar los que tienen mas probabilidad de continuar la secuencia
 - Durante el entrenamiento fue como afinar cada tecla del piano y ajustar cada cuerda. Ahora cuando “tocas” una secuencia de teclas (escribes), el piano ya sabe qué sonidos hacer porque ya está todo afinado.
- **Attention mechanism**
 - Es como cuando lees un cuento y prestas más atención a las partes importantes. Los GPT hacen lo mismo con las palabras: ponen más atención a las palabras **que creen que son más** importantes de tu pregunta para darte una “**mejor**” respuesta.



- En el chatbot TÚ eres quien habla con el LLM (modelo de lenguaje). En la API, es tu PROGRAMA quien habla con el LMM, que hace las tareas automáticamente sin que tú tengas que estar ahí escribiendo cada vez.

Buenas practicas de IA: 1º “qué” y “por qué”, sólo después uso la IA si me puede ayudar. Las herramientas no son “buenas o malas” es el uso que hacemos (adecuado/no adecuado)



Main learning outcomes of the first part:

- Critically assess a set of recommendations for the use of AI for research.
- Be able to choose which activities you are interested in delegating or co-creating with IAgén
 - To do this, you must know the VA model applied to IAgén

Take aways first part:

- Riesgo de que acabemos teniendo una investigación trivial y superficial que impida un aprendizaje profundo.
 - Podemos quedarnos anclados en investigar sólo el tipo de preguntas que los GPT puedan responder
 - El esfuerzo y el tiempo dedicado es parte del aprendizaje
 - ¿Queremos aprender nosotros? (y retener actualizado ese aprendizaje)
 - ¿Nos basta que aprenda la máquina?
- Primero hay que pensar qué necesitamos y el motivo por el que lo necesitamos
- Luego vemos si la IA nos puede ayudar a hacer **parte** de eso mejor o más rápido
 - Y el resto de cosas, las hacemos “a mano” como siempre
 - Pero no las dejamos de hacer



Learning outcomes for the next part

- Remember the steps of a systematic review. Especially up to the screening phase
- Apply 3 modes (manual, **prompt**, embeddings proximity) to screening phase in two simplified cases
 - A common case. Umbrella reviews for "Student evaluation of teaching"
 - A specific case for each participant
- Learn about the supervised learning option (rayyan vs Python code). We will only comment on it as a possibility, we will not go into this now.
- De todas las cosas que se pueden hacer con IAgén, en la segunda parte me voy a centrar en el caso de uso de una SLR (solo los pasos iniciales hasta cribado)
 - No tenéis por qué usar IA para esto, es solo un ejemplo de uso
 - Podéis extrapolar alguno de estos ejemplos a otras tareas de investigación

Somos la última generación que sabrá hacer SLR y otras muchas cosas... y eso nos hará imprescindibles



- Aunque exista la IAgén necesitamos aprender a hacer SLR (algunas personas).
- En unos ¿10 años? Todo el proceso estará automatizado (búsqueda, filtrado, codificación o extracción de tabla de datos, análisis y resumen de conclusiones)

Lo están intentando, pero de momento no les sale bien (o al menos no siempre o en todas las disciplinas). Ejemplo de Consensus (julio 2025)

SLR SET effect on TEachers

1 query

Share

****ROL Y OBJETIVO**** Actúa como un investigador especializado en psicología educativa con experiencia en riesgos psicosociales laborales. Tu objetivo es realizar una revisión sistemática para evaluar el impacto en la salud mental del profesorado universitario derivado de leer comentarios negativos estudiantiles, destinada a informar decisiones de gestión al equipo directivo universitario para un artículo de investigación. ****CONTEXTO PROFESIONAL**** Esta revisión será utilizada por el equipo directivo de la universidad para desarrollar políticas de apoyo al profesorado y protocolos de gestión de evaluaciones estudiantiles. ****MARCOS TEÓRICOS DE REFERENCIA**** Fundamenta tu análisis en: - Modelo Demanda-Control-Apoyo de Karasek (evaluando cómo los comentarios incrementan demanda psicológica sin aumentar control) - Modelo Esfuerzo-Recompensa de Siegrist (analizando desequilibrios entre esfuerzo docente y recompensas percibidas) ****INSTRUCCIONES ESPECÍFICAS**** 1. ****Pregunta de investigación****: "¿Cómo afecta, o podría afectar, desde el punto de vista de riesgos psicosociales, a un profesor universitario el leer los textos que hayan escrito sus estudiantes como respuesta a una pregunta abierta sobre aspectos que no les gustan de las asignaturas?" 2. ****Población objetivo****: Profesorado universitario en general (sin restricciones por experiencia, disciplina o institución) 3. ****Alcance temporal****: Sin restricción temporal 4. ****Metodología requerida****: Marco de trabajo PRISMA o Cochrane ****COMPONENTES A DESARROLLAR**** ****Protocolo de revisión sistemática que incluya****: - Estrategia de búsqueda detallada con términos MeSH específicos - Criterios de inclusión/exclusión explícitos - Bases de datos a consultar - Procedimientos de selección y extracción

Ask a follow up...

Deep Search BETA

****ROL Y OBJETIVO**** Actúa como un investigador especializado en...

748 Identified | 639 Screened | 251 Eligible | 0 Included

- Initial Survey & Search Strategy
1 search · 0 new eligible papers
- Break down query
3 searches · 40 new eligible papers
- Rephrase terminology
4 searches · 53 new eligible papers
- Zoom in on models
3 searches · 115 new eligible papers
- Identify contrasting findings
4 searches · 9 new eligible papers
- Expand to related risks
 - burnout and mental health among university faculty
 - emotional labor and psychosocial risks in higher education teaching staff
 - job satisfaction and stressors in academic work environment
- Expand to adjacent psychosocial risks in academic work (e.g., burnout, emotional labor, job satisfaction) to contextualize findings.
3 searches · 34 new eligible papers
- Find foundational work
3 searches · 1 filter · 0 new eligible papers
- Ranking papers

Could not find any papers directly addressing how reading negative student comments impacts university faculty mental health. Consider broadening your search or removing some filters.

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SINCOS

74

En este caso, menos es mas https://consensus.app/search/pueden-afectar-negativamente-a-los-profesores-univ/3YyZbh7IQHWvPmaV_sMK5g/?utm_source=share&utm_medium=clipboard

¿pueden afectar negativamente a los profes...

1 query

Share

¿pueden afectar negativamente a los profesores universitarios el leer las preguntas de texto abierto en las Student evaluation of teaching surveys?

Deep · Step 2 / 8 · Zoom in on open-ended

Filter

...

Results

Sexism, racism, prejudice, and bias: a literature review and synthesis of research surrounding student evaluations of courses and teaching

Student evaluations are influenced by racist, sexist, and homophobic prejudices, and are biased against discipline and subject area.

Literature Review Highly Cited

2021 · 148 citations · Troy A. Heffernan · Assessment & Evaluation in Higher Education

Contribution of open-ended questions in student evaluation of teaching

Open-ended questions in student evaluation of teaching surveys contribute significantly to the analysis of professors' teaching performance, providing concrete feedback and actionable topics for improving teaching effectiveness.

Observational Study

Ask a follow up...

Pro

Filter

Deep Search BETA

¿pueden afectar negativamente a los profesores universit

50 Identified

50 Screened

50 Eligible

- Incl

Initial Survey & Search Strategy

1 search · 50 new eligible papers

Zoom in on open-ended

Impact of reading open-ended student comments on university professors

Emotional effects of open-ended feedb evaluations on faculty

Negative consequences of qualitative r student teaching evaluations

Focus on the specific impact of reading opi responses in student evaluations on univers

3 searches · - new eligible papers

Rephrase for terminology

4 searches · - new eligible papers

Identify negative effects

3 searches · - new eligible papers

Contrast with positive effects

4 searches · - new eligible papers

Explore adjacent constructs

3 searches · - new eligible papers

Find foundational work

3 searches · 1 filter · - new eligible papers

Ranking papers

¿pueden afectar negativamente a los profesores universitarios el leer las preguntas de texto abierto en las Student evaluation of teaching surveys?

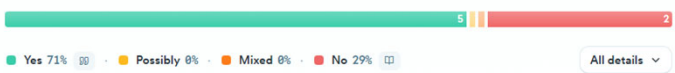
Deep · 21 searches · 50 sources

Reading open-ended questions in student evaluation of teaching surveys can negatively affect university professors, especially through exposure to abusive, biased, or prejudiced comments, which may harm their well-being and professional advancement.

1. Introduction

Student Evaluation of Teaching (SET) surveys are a ubiquitous tool in higher education, often including open-ended questions that allow students to provide qualitative feedback on their instructors. While these comments can offer valuable insights for teaching improvement, a growing body of research highlights significant negative effects on university professors, particularly when comments are abusive, biased, or discriminatory. Studies have shown that open-ended responses can be a source of stress, anxiety, and even harm to faculty well-being, with women and marginalized groups disproportionately targeted by derogatory remarks 1 32 30 27 28 26 17. The prevalence of such comments, combined with the use of SETs in high-stakes decisions like promotion and tenure, raises concerns about the fairness and validity of these evaluations 1 32 27 28. Furthermore, the emotional toll of reading negative or abusive feedback can impact professors' mental health, job satisfaction, and teaching practices 30 26 17. This review synthesizes the current literature on the negative impacts of reading open-ended SET comments on university professors, with a focus on the types of harm experienced, the groups most affected, and the broader implications for academic policy and practice.

Do open-ended survey responses negatively affect university professors? N = 7



2. Methods

A comprehensive search was conducted across over 170 million research papers in Consensus, encompassing databases such as Semantic Scholar and PubMed. The search strategy included multiple targeted queries related to the impact of open-ended questions in SETs on university professors, focusing on emotional, psychological, and professional consequences. In total, 1,043 papers were identified, 668 were screened, 477 were deemed eligible, and the 50 most relevant papers were included in this review.

Search Strategy



FIGURE 1 Flow diagram of paper selection for this review.



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Con scispace <https://scispace.com/search/pueden-afectar-negativamente-a-los-profesores-universitarios-8dwyjq4clc?q=psychological+impact+of+reading+open-ended+questions+in+student+evaluations+on+university+professors>

SCISPAC

Search or ask a question

mental health or stress levels of professors? (e.g., anxiety, motivation, job satisfaction)

2. **Professional Development:** Would you like to focus on how these evaluations influence the professional growth or teaching methods of professors? (e.g., changes in teaching style, professional training)

3. **Institutional Policies:** Are you curious about how universities might use the feedback from these surveys to make policy or administrative decisions affecting professors? (e.g., tenure decisions, performance reviews)

4. **Comparative Analysis:** Should the research compare the effects on professors across different departments or institutions? (e.g., humanities vs. sciences, public vs. private universities)

Feel free to select one or more options, or provide additional details so I can assist you better with your search.

psychological impact

Performing Steps (takes upto 3 minutes) 30% Done

Executing multiple relevant queries to broaden the search.

psychological impact of reading open-ended questions in student evaluations on university professors

impact of qualitative student feedback on teacher psychology and emotional well-being in higher education

effects of student feedback on professors' emotional health and professional growth in higher education

Finding relevant papers for each query with applied filters

46 top papers found out of 1050

[1] Teacher Wellbeing: The Importance of Teacher-Student Relationships
Jantine L. Spitt + 2 more - Educational Psychology Review

[2] University lecturers' emotional responses to and coping with student feedback: a Finnish case study
Sonja Lutovac + 3 more - European Journal of Psychology of Education

[3] Nurturing learning development through student feedback
Colette Mair + 4 more - Journal of Learning Development in Higher Education

Show More

Performing Steps (takes upto 3 minutes) 80% Done

Executing multiple relevant queries to broaden the search.

psychological impact of reading open-ended questions in student evaluations on university professors

impact of qualitative student feedback on teacher psychology and emotional well-being in higher education

effects of student feedback on professors' emotional health and professional growth in higher education

Emotional and psychological effects of anonymous student evaluations on faculty wellbeing in higher education

emotional intelligence and coping mechanisms of university professors in response to qualitative student feedback

Finding relevant papers for each query with applied filters

91 top papers found out of 1750

[1] Distinguishing the Themes Emerging from Masses of Open Student Feedback
Timo Hynninen + 3 more - International Convention on Information and Communication Technology, Electronics and Microelectronics

[2] Contribution of open-ended questions in student evaluation of teaching
Pablo Marshall - Higher Education Research & Development

[3] Feedback as Open-Ended Conversation: Inviting Students to Co-Regulate and Metacognitively Reflect During Assessment
Jt Torres - Journal of the Scholarship of Teaching and Learning

Show More

Reviewing citations & references to identify additional relevant works

99 relevant citations and references found

[1] We can work it out: faculty interpretation of student evaluations of teaching
Yali Bai - Journal of Further and Higher Education

[2] The relationship between students' evaluations of teaching and academics professional development
Mari Karm + 2 more - Journal of Further and Higher Education

SCISPAC

Search or ask a question

My Library / Notebooks / Student evaluation of teaching

File Format Tools

Psychological Impact of Reading Open-Ended Questions in Student Evaluations on University Professors

Introduction

Student evaluations of teaching (SET) are a common practice in higher education, intended to improve teaching quality and provide insights into the learning experience. However, the psychological impact of reading open-ended questions in these evaluations on university professors has become a significant concern. This response explores the emotional, professional, and institutional implications of such feedback, drawing on insights from relevant research papers.

Emotional Impact of Open-Ended Feedback

The emotional impact of open-ended feedback in SET is profound. Professors often experience negative emotions such as sadness, anxiety, and stress when receiving negative feedback, particularly when it pertains to their teaching competence (Mundtshausen et al., 2024). These emotions can be exacerbated by the anonymous nature of the feedback, which sometimes includes hurtful, defamatory, or abusive comments (Cunningham et al., 2023; Bivas, Detsios, Dotsier and Coping: The Impact of Anonymous Student Evaluation of Teaching on the Health of Higher Education Teachers, 2023). The emotional toll is further compounded by the perception that such feedback is beyond their control and may be unfair or biased (Mundtshausen et al., 2024; Plachemov et al., 2023). Younger academics and those in non-tenured positions are particularly vulnerable to

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SINCOE

76

Con scite, localiza un trabajo mio y dice que digo algo que jamás dije.. (solo es cuestión de tiempo que mejoren los modelos para que no metan la pata así, porque es trivial mejorarlos –pero cuesta dinero-)

scite_

AssistantProduct Solutions

Ask me another questionAsk

Custom Assistant Settings are being used for this session.

¿pueden afectar negativamente a los profesores universitarios el leer las preguntas de texto abierto en las Student evaluation of teaching surveys?

Los profesores universitarios se enfrentan a una serie de desafíos inherentes a las evaluaciones de enseñanza por parte de los estudiantes, especialmente cuando se trata de preguntas de texto abierto en las encuestas de evaluación. Estas preguntas ofrecen la oportunidad de obtener retroalimentación más matizada y rica en detalles sobre la experiencia del estudiante, pero también pueden presentar varios problemas que podrían afectar negativamente a los docentes.

Primero, la retroalimentación abierta puede conducir a la interpretación errónea de las intenciones del instructor. Según Torres (Torres, 2022), dejar a los estudiantes interpretar las intenciones de los profesores puede dificultar el aprendizaje, sugiriendo que las respuestas abiertas pueden perjudicar la autoestima profesional de los docentes si se perciben como críticas contundentes o mal dirigidas. Estudios han mostrado que estas evaluaciones pueden influir en la moral del profesorado, ya que la naturaleza subjetiva de la retroalimentación permite que comentarios negativos sean particularmente impactantes (Aznar-Mas et al., 2023).

Además, las evaluaciones al final del semestre, como las encuestas de evaluación de enseñanza, suelen no proporcionar retroalimentación a tiempo para que los instructores hagan ajustes en su enseñanza. Chang et al. indican que estas evaluaciones son a menudo demasiado tardías para beneficiar a los docentes en el mismo curso, lo cual se traduce en una percepción de que el tiempo y esfuerzo invertido en la recopilación de retroalimentación es, en muchos casos, infructuoso (Chang et al., 2014). Esto puede desincentivar a los profesores a involucrarse con los procesos de evaluación, viéndolos como una formalidad más que como una herramienta de desarrollo profesional.

Otro aspecto a considerar es el sesgo que puede surgir de las respuestas en las encuestas de evaluación de enseñanza. Un análisis de la literatura indica que las evaluaciones de estudiantes a menudo se ven afectadas por preconcepciones y sesgos inherentes, lo que puede llevar a que comentarios poco constructivos alteren la percepción general sobre la eficacia docente (Hagen, 2020). Esta percepción sesgada es problemática en el contexto de educación superior, donde la reputación del profesor puede impactar directamente en su futuro profesional y en la dinámica del aula.

La diversidad de comentarios también puede ser abrumadora para los docentes. Grönberg et al. (Grönberg et al., 2021) mencionan que, aunque las preguntas abiertas deberían proporcionar información valiosa, a menudo se limitan a simples descripciones que no son analizadas adecuadamente, dejando a los profesores a navegar por un mar de información que puede ser difícil de sintetizar y aplicar constructivamente. Esto puede resultar en una carga cognitiva adicional para los educadores, quienes deben discernir qué comentarios son relevantes.

Asimismo, la reducción de la calidad del feedback puede ser atribuida a la falta de análisis detallado de las respuestas abiertas, que frecuentemente son ignoradas en favor de las respuestas cuantitativas más fácilmente evaluables (Nawaz et al., 2022). Este enfoque puede socavar el potencial de la retroalimentación cualitativa, llevándola a ser

ReferencesSearch Strategy

"...Open-ended feedback, however, also has its limits. Leaving students to interpret teachers' intentions can not only hinder lea...">>

Section: Discussion

SaveView PDF

Cite

Effectiveness of the use of open-ended questions in student evaluation of teaching in an engineering degree

Lourdes E. Aznar-Mas¹, Lorena Atarés², Juan A. Marín-García³ 2023

JTEM

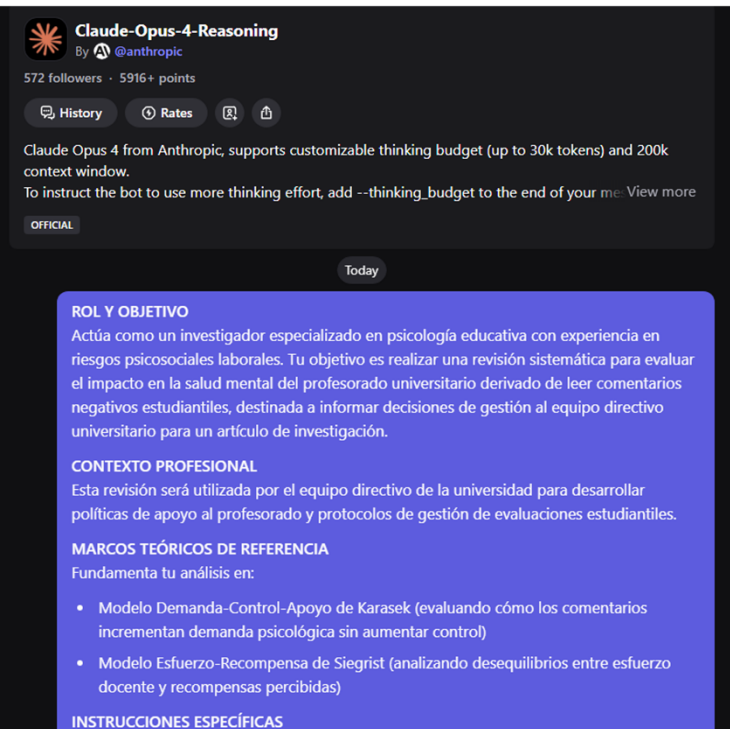
2000000

"...Student evaluation of teaching is the systematic process in which school leaders periodically critique teachers' work performance based on student feedback (Lejonberg, Elstad & Christophersen, 2018) . Although initially considered a highly controversial topic, teaching evaluation has long been seen as an integral part of good professional practice in Higher Education institutions (Hounsell, 2003)..."<<

Section: Introduction

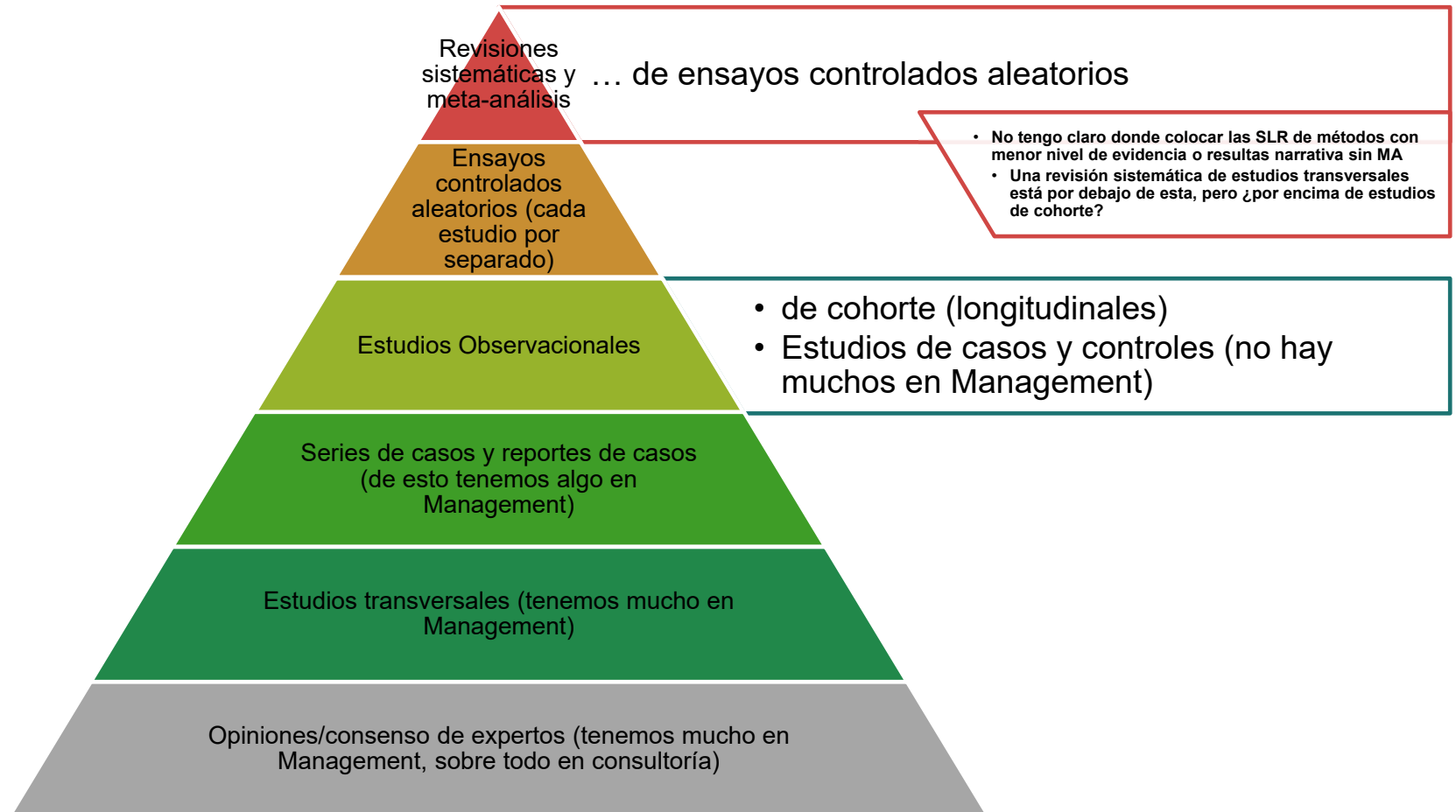
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Con otras herramientas con el prompt largo

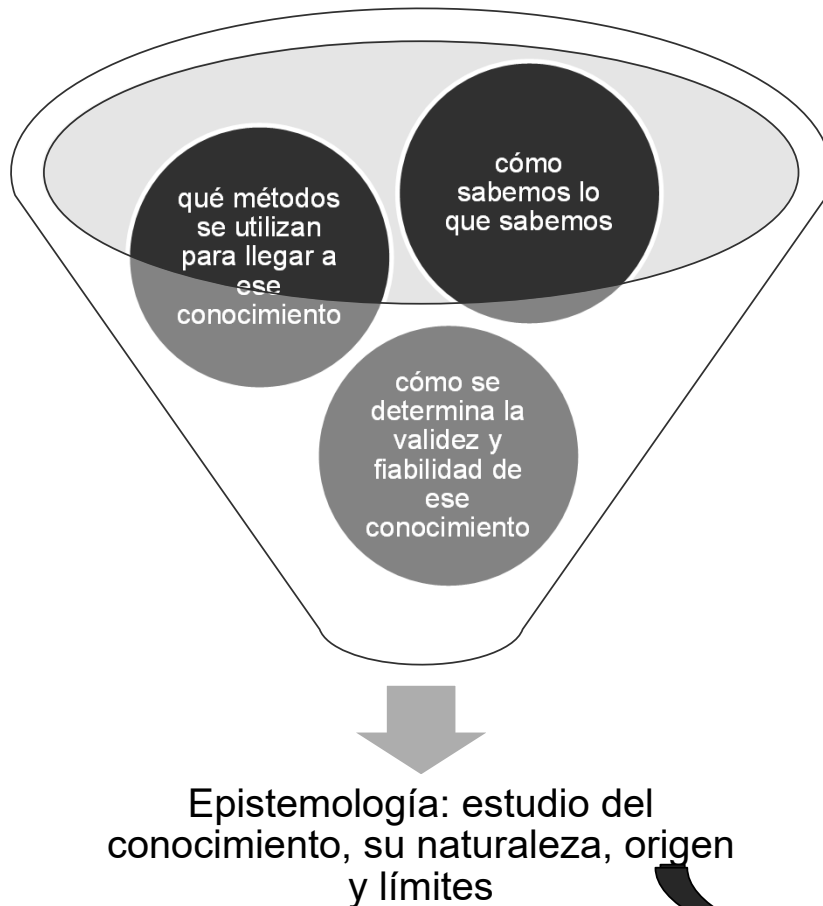


- Perplexity
 - <https://www.perplexity.ai/search/rol-y-objetivo-actua-como-un-i-umwUbSh3TfSbnQjERI4nJw>
- ChatGPT 4.5
 - [Revisión riesgos psicosociales docentes](#)
- Claude-sonnet-4
 - https://poe.com/s/24JH28KZtOEQL8b0bLy_m

Desde el punto de vista positivista. La síntesis de evidencias está en la cúspide de la pirámide de evidencia científica (aplica a MA, no se si a todas las SLR)



Reflexi3n epistemol3gica: lo anterior es cierto desde una visi3n positivista de la ciencia



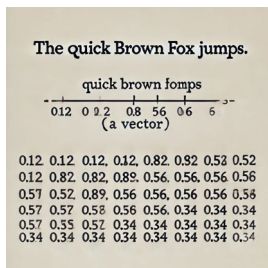
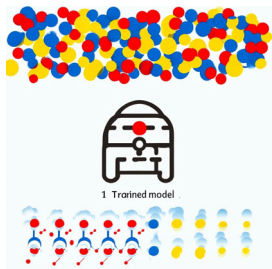
Enfoque interpretativo o constructivista:

1. Se valoran m3s los métodos cualitativos como estudios de caso en profundidad, etnografías, o investigación-acci3n.
2. La validez se basa m3s en la riqueza y profundidad de los datos que en la generalizaci3n estadística.

Perspectiva crítica:

1. Se priorizan métodos que revelan estructuras de poder y promueven el cambio social.
2. La investigaci3n participativa y la investigaci3n-acci3n crítica son altamente valoradas.

Terminology: four types of AI that I am going to use. The accompanying last name is very relevant.



- Generative AI
 - Generalist generative (example Claude4.SONNET; GPT4O [GPTO1]; Gemini 1.5...)
 - Generative [supposedly] trained in Research (perplexity, scite, consensus, ...)
- AI with natural language processing (NLP)
 - Sentence transformers
- AI [traditional] supervised learning (ML)
 - (Rayyan as an example)

Etapas de una Revisión Sistemática de Literatura. Solo voy ha hacer el caso de uso HASTA screening (creo que es una tarea muy costosa con poco valor añadido... tiene valor añadido las primeras 100-300 iteraciones. Pero 10.000 haría inviable y no “reliable” el procesos con humanos)

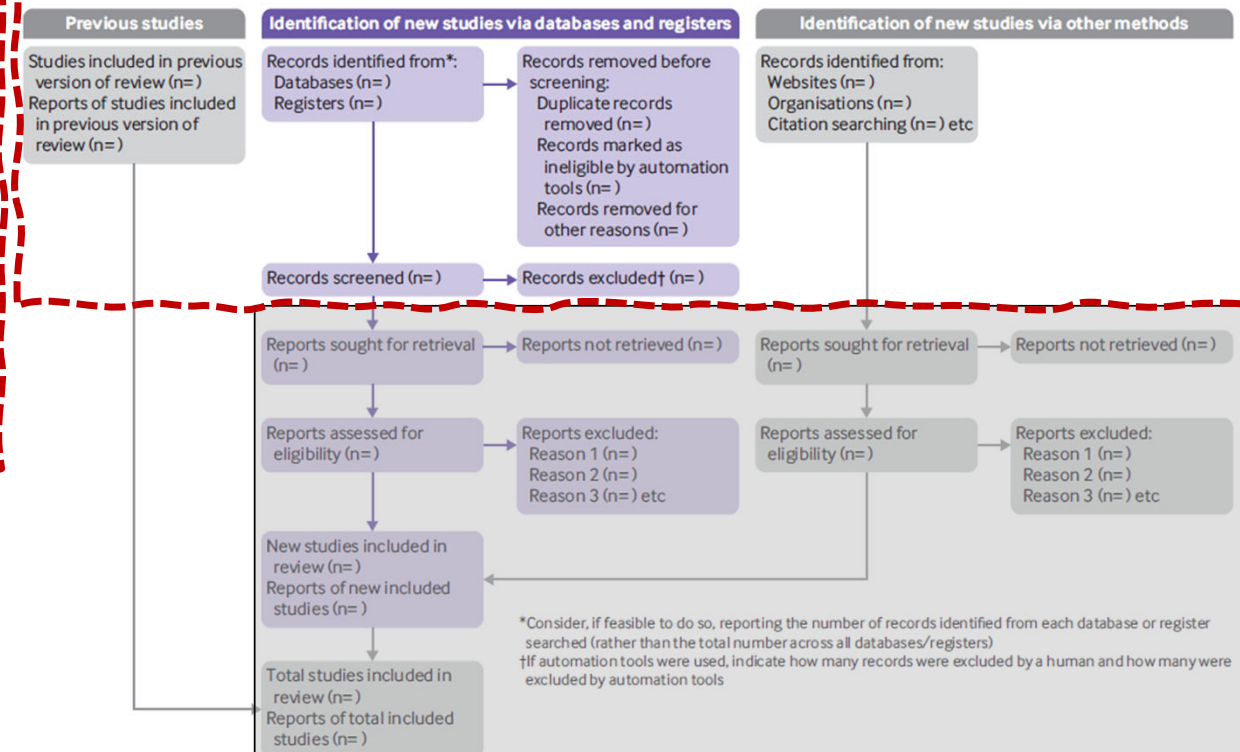
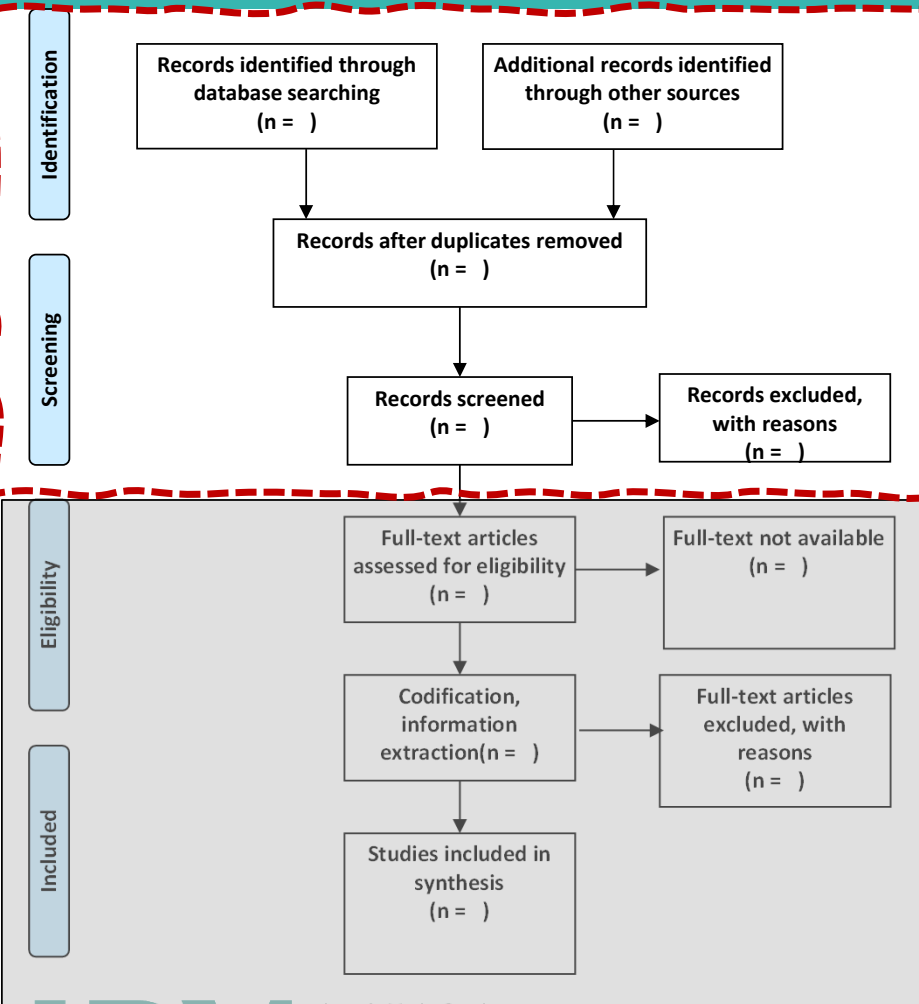
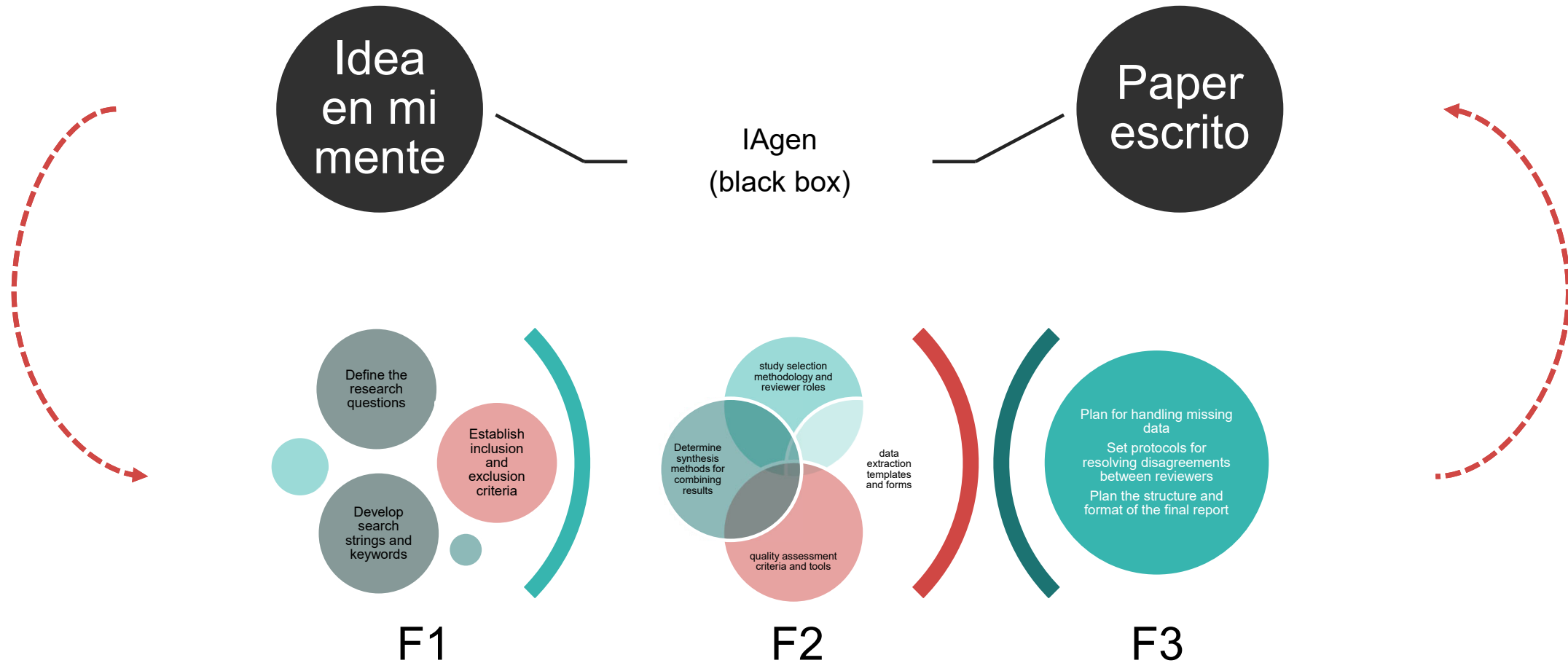


Fig 1 | PRISMA 2020 flow diagram template for systematic reviews. The new design is adapted from flow diagrams proposed by Boers,⁵⁵ Mayo-Wilson et al.⁵⁶ and Stovold et al.⁵⁷ The boxes in grey should only be completed if applicable; otherwise they should be removed from the flow diagram. Note that a “report” could be a journal article, preprint, conference abstract, study register entry, clinical study report, dissertation, unpublished manuscript, government report or any other document providing relevant information.

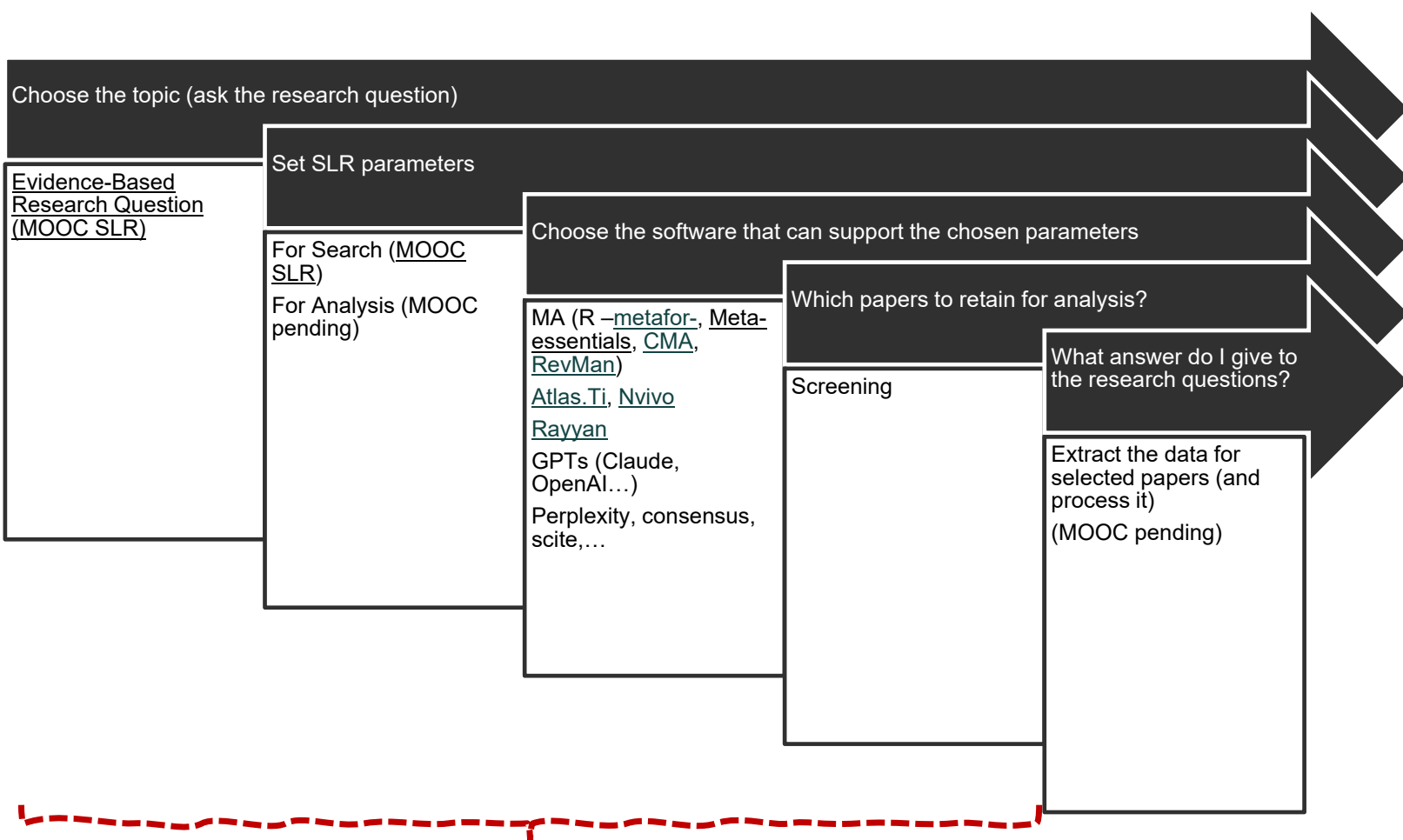
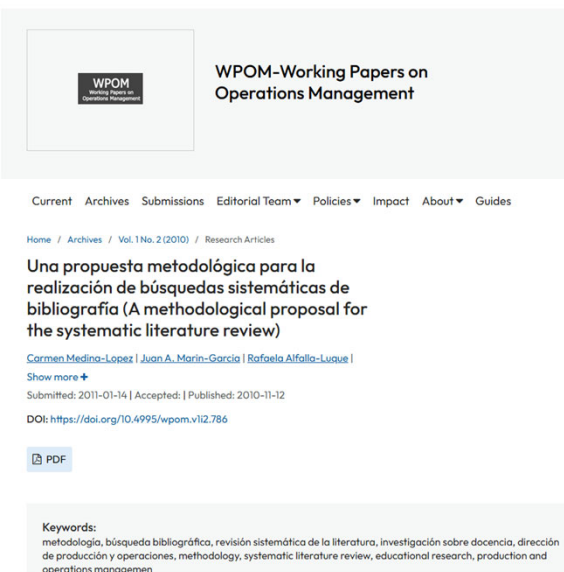
thebmj | BMJ 2021;372:n71 | doi:10.1136/bmj.n71

¿Existe la magia?



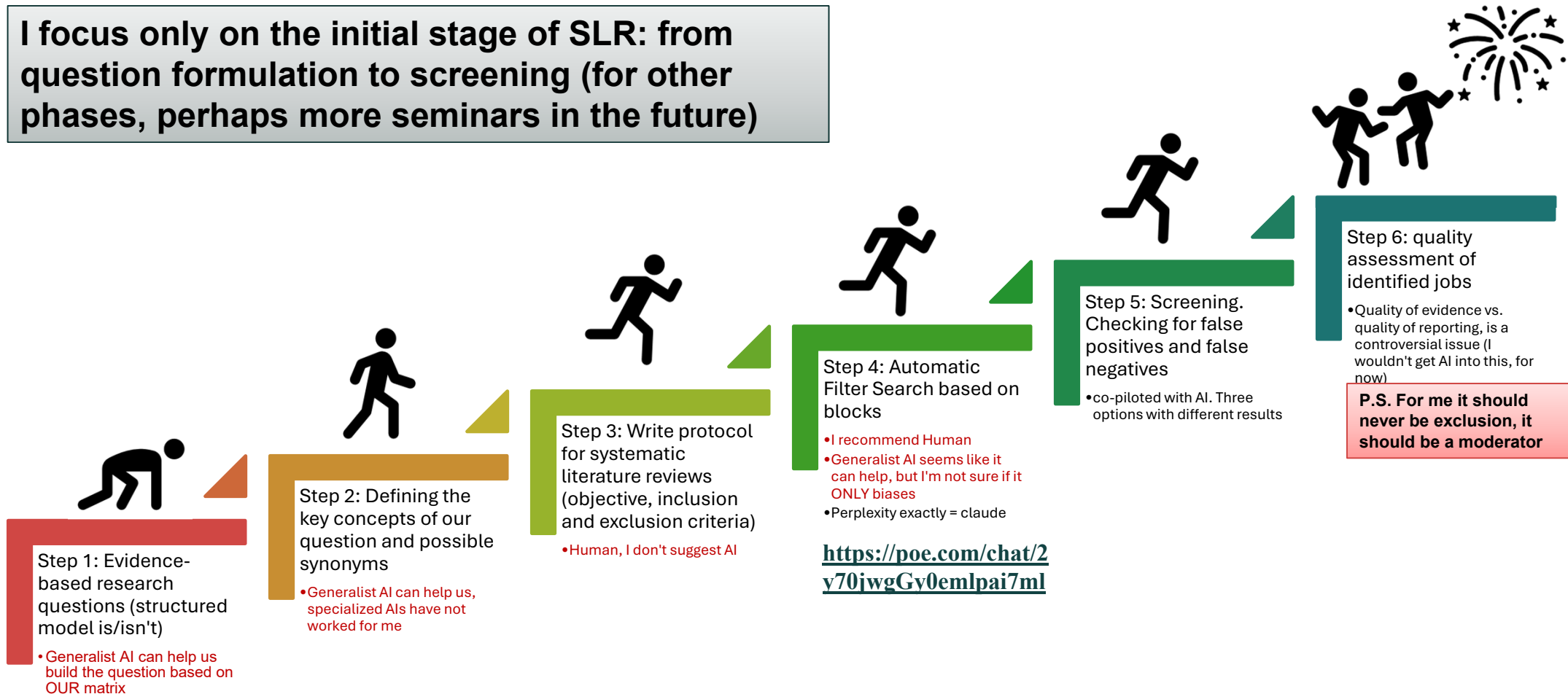
Basic decisions in a systematic literature review

Medina-Lopez, C., Marin-Garcia, J. A., & Alfalla-Luque, R. (2010). Una propuesta metodol3gica para la realizaci3n de b3squedas sistemáticas de bibliografía (A methodological proposal for the systematic literature review). *WPOM-Working Papers on Operations Management*, 1(2), 13–30.
<https://doi.org/10.4995/wpom.v1i2.786>



Summary of the session. The rest of the time I will dedicate it to show examples of support to the statements of this transparency

I focus only on the initial stage of SLR: from question formulation to screening (for other phases, perhaps more seminars in the future)



Step 1: Evidence -based research questions (structured model is/not). The AI generalist can help us build a title based on our matrix

	Is	It's not
What is the population or characteristics of the people who are going to be used or will the decision or intervention be affected?	University Professors	Students, Service Staff Other educational levels other than university
[In the case of interventions] What is the intervention, program, treatment, problem, improvement, product, service... to analyze?	Open-ended questions to capture students' opinion in the "students evaluation of teaching"	Closed-ended likert scale questions or similar
[In other cases] why is the information, or the decision, necessary? What reasons drive the behavior or experience?	It is possible that when reading the opinions of the students, teachers may feel offended, insulted or simply questioned	The usefulness of this information for managers/heads of university, centre, department, degree or subject
What are the expected results, expectations, impact and how to measure them?	The impact on the psychological health, well-being or motivation of teachers	The quality of teaching or learning
Who are the people who are going to provide the service or information or are going to perform the intervention	Responsible for teaching quality in universities	
Tipo de revisión [critical, descriptive, qualitative systematic, scoping, meta-analysis, umbrella]	Umbrella	To analyse the primary studies, I want to review the results of the reviews
[if any] What is the group against which we are going to compare the results?		



Title: **"The Impact of Open-ended Student Teaching Evaluations on University Professors' Psychological Health and Well-being: An Umbrella Review "**

Step 1: Evidence -based research questions (structured model is/not). The AI generalist can help us build a title based on our matrix

<https://www.youtube.com/watch?v=cj6cvI9HBEM>

Título Pico Spider eclipse Cochrane - Poe



PROMPT (claude3.7-sonnet); Based on the information in the attached table, propose a title for a systematic literature review that follows the recommendations of Cochrane Reviews (with the PICO - SPIDER-ECLIPSE framework)."

Based on the information provided in the table, I can see this pertains to a systematic review about the impact of student evaluations of teaching on university professors' psychological health and well-being. Using the SPIDER framework (which is often more appropriate than PICO for qualitative reviews):

S (Sample): University Professors

P (Phenomenon of Interest): Open-ended student evaluations of teaching

I (Design): Not specifically mentioned, but appears to be reviewing literature on psychological impacts

E (Evaluation): Impact on psychological health, well-being, and motivation

R (Research type): Umbrella review (as indicated in the table)

Here's a proposed title for the systematic review following Cochrane recommendations:

"The Impact of Open-ended Student Teaching Evaluations on University Professors' Psychological Health and Well-being: An Umbrella Review"

Alternative title using the ECLIPSE framework elements:

"Psychological Effects of Student Feedback Systems in Higher Education: An Umbrella Review of Open-ended Teaching Evaluations and Their Impact on University Professors' Well-being"

With Gemini2.5 Free

	It is	It's not
Open-ended questions to capture students' opinion in the evaluation of teaching	Yes	No
Closed-ended Likert scale questions	No	Yes
Qualitative student feedback	Yes	No
Quantitative measures of student feedback	No	Yes
Qualitative student feedback from teaching evaluations	Yes	No
Quantitative measures of student feedback from teaching evaluations	No	Yes
Qualitative student feedback from teaching evaluations	Yes	No
Quantitative measures of student feedback from teaching evaluations	No	Yes

Aunque te escriba en castellano mis prompts, responde siempre en inglés, empezando la respuesta siempre con la traducción de mi prompt a inglés. Prompt: "partiendo d..."

Mostrar cuando piensa

Okay, here's a proposed title for your systematic literature review, based on the information you provided and following Cochrane Review recommendations.

User Prompt Translation: "Based on the information in the attached table, propose a title for a systematic literature review that follows the recommendations of Cochrane Reviews (with the PICO - SPIDER - ECLIPSE framework)."

Proposed Title for the Systematic Literature Review:

"The impact of qualitative student feedback from teaching evaluations on the psychological well-being of university professors: An umbrella review"

Explanation and Alignment with Frameworks:

This title aims to be clear, specific, and encompass the key elements outlined in your table, fitting the structure often recommended for systematic reviews, including those by Cochrane. An umbrella review synthesizes evidence from multiple systematic reviews.

Here's how the title aligns with the PICO framework (though umbrella reviews sometimes have a broader scope than a single PICO question, the elements are still relevant):

- P (Population): University professors**
 - Your table specifies "University Professors."
- I (Intervention/Exposure): Qualitative student feedback from teaching evaluations**
 - This corresponds to "Open-ended questions to capture students' opinion in the 'students evaluation of teaching'" from your table. "Qualitative student feedback" accurately represents this type of input, differentiating it from quantitative measures (which your table lists under "It's not" – e.g., "Closed-ended Likert scale questions").
- C (Comparison):** (Not explicitly stated in this title, which is common for umbrella reviews focusing on an exposure and its outcomes).
 - For an umbrella review, the comparison is often implicit in the scope of the included systematic reviews (e.g., reviews that examine this type of feedback versus those that don't, or reviews looking at different types of impact). If a direct comparison were the focus of the primary reviews being synthesized, it could be added, but the current information suggests a focus on the impact of a specific type of feedback.

- **O (Outcome): Psychological well-being**
 - This encompasses "The impact on the psychological health, well-being or motivation of teachers" specified in your table. "Psychological well-being" is a comprehensive term that covers these aspects.

Alignment with SPIDER framework (often used for qualitative or mixed-methods reviews, relevant here due to the nature of the feedback and the umbrella review of potentially diverse studies):

- **S (Sample): University professors**
- **PI (Phenomenon of Interest):** The experience and **impact of qualitative student feedback** from teaching evaluations, specifically focusing on **psychological well-being**.
- **D (Design):** The design of the review itself is an **umbrella review**. The design of studies within the included systematic reviews will vary.
- **E (Evaluation):** The **psychological well-being** (health, motivation) of professors.
- **R (Research Type): Umbrella review** (review of systematic reviews), as explicitly stated in the title and your table ("Tipo de revisión: Umbrella").

Alignment with ECLIPSE framework (often used for policy or service-related reviews):

- **E (Expectation):** To understand the **impact...on psychological well-being**.
- **C (Client group): University professors** (as the group affected).
- **L (Location):** Implied as **universities** (where teaching evaluations occur for professors).
- **I (Impact): Psychological well-being**.
- **P (Professionals): University professors**.
- **S (Service):** The process of **student evaluations of teaching** (specifically qualitative feedback aspects).

Cochrane Review Recommendations:

Cochrane guidance generally suggests titles should be concise, precise, and clearly reflect the scope of the review, often including key PICO elements for intervention reviews. For overviews of reviews (umbrella reviews), it's crucial to indicate this in the title. This proposed title clearly states "An umbrella review."

This title clearly delineates the population, the specific type of exposure (qualitative feedback from student evaluations), the primary outcome of interest (psychological well-being), and the nature of the review (umbrella review), aligning with best practices for systematic review titles.

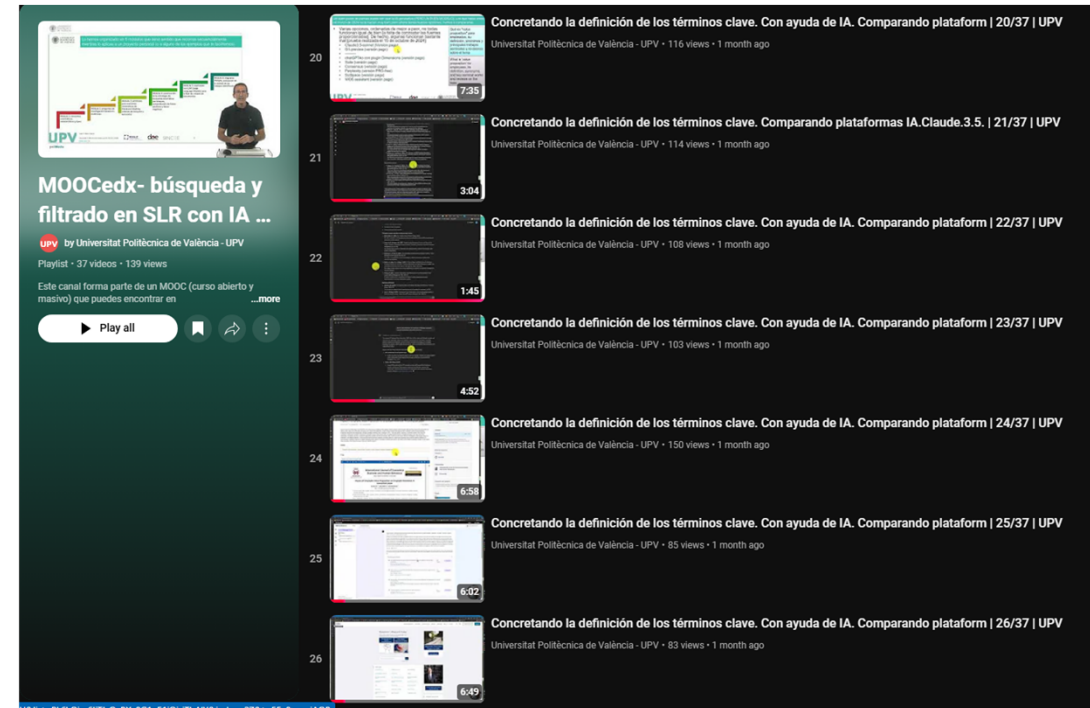
Step 2: Definition of the key concepts of our question and possible synonyms. The generalist AI can help us, the specialized ones have not worked for me (in January 2025)



- Support videos (in Spanish). Looking for definitions, synonyms and seminal works with 8 generative AI platforms (they are part of a MOOC but you do not need to do the MOOC if you only are interested in this)

<https://www.youtube.com/playlist?list=PL6kQim6ljTJsQvBXx8G1nE1iOi-iTb4JY>

- Claude3.5-sonnet (Version pago)
- O1-preview (versión pago)
- -----
- chatGPT4o con plugin Dimensions (versión pago)
- Scite (versión pago)
- Consensus (versión pago)
- Perplexity (versión PRO-free)
- SciSpace (versión pago)
- WOS assistant (versión pago)



I will show a result with Claude3.7sonnet. Test it (with my prompt, or the one that interests you) and compares with Gemini 1.5 o Chat GPT



<https://g.co/gemini/share/644b9dbd24c0>

- Why Gemini1.5?
Because it is free and free access (with Gmail account)
- There is also the issue of privacy ... everything you upload to Gemini is used to train ... and for other things

- "What is 'Student evaluation of teaching', its definition, synonyms, and key seminal works and reviews on the topic"

Save the "elaboration" because we will use it later

Hay revisores humanos que leen algunas conversaciones guardadas para mejorar la IA de Google. Para evitar que esto suceda en conversaciones futuras, desactiva el ajuste Actividad en las aplicaciones de Gemini. Si este ajuste está activado, no introduces información que no quieras que se revise ni use. [Cómo funciona](#)

"What is 'Student evaluation of teaching', its definition, synonyms, and key seminal works and reviews on the topic" Claude 3.7

Definition

Student Evaluation of Teaching (SET) refers to the systematic process of collecting feedback from students about their educational experiences, instructor effectiveness, and course quality. These evaluations typically occur at the end of a course or academic term and involve students rating various aspects of teaching performance and course design through questionnaires or surveys. SETs are used for various purposes including:

- Providing feedback to instructors for improvement
- Administrative decision-making (promotion, tenure, merit pay)
- Course development and curriculum enhancement
- Quality assurance in higher education
- Research on teaching effectiveness

Synonyms and Related Terms

- Course evaluations
- Teaching evaluations
- Student ratings of instruction (SRI)
- Student feedback questionnaires
- Student assessment of teaching (SAT)
- Student perception of teaching (SPOT)
- Student evaluations of educational quality (SEEQ)
- Student ratings of teaching effectiveness (SRTE)
- Student course experience questionnaires (SCEQ)
- Teaching effectiveness measures

Key Seminal Works and Reviews

Foundational Works

1. **Marsh, H. W. (1984).** "Students' evaluations of university teaching: Dimensionality, reliability, validity, potential biases, and utility." *Journal of Educational Psychology*, 76(5), 707-754.
 1. One of the most cited works establishing the psychometric properties of SETs
2. **Feldman, K. A. (1989).** "The association between student ratings of specific instructional dimensions and student achievement." *Research in Higher Education*, 30(6), 583-645.
 1. Connected specific teaching behaviors to student outcomes
3. **Cohen, P. A. (1981).** "Student ratings of instruction and student achievement: A meta-analysis of multisection validity studies." *Review of Educational Research*, 51(3), 281-309.
 1. Early meta-analysis showing moderate correlations between ratings and achievement

Influential Reviews and Meta-analyses

4. **Spooren, P., Brockx, B., & Mortelmans, D. (2013).** "On the validity of student evaluation of teaching: The state of the art." *Review of Educational Research*, 83(4), 598-642.
 4. Comprehensive review of validity issues in SETs
5. **Uttl, B., White, C. A., & Gonzalez, D. W. (2017).** "Meta-analysis of faculty's teaching effectiveness: Student evaluation of teaching ratings and student learning are not related." *Studies in Educational Evaluation*, 54, 22-42.
 4. Challenged previous findings on the relationship between SETs and learning
6. **Hornstein, H. A. (2017).** "Student evaluations of teaching are an inadequate assessment tool for evaluating faculty performance." *Cogent Education*, 4(1).
 4. Critical analysis of SETs as assessment tools

Bias and Criticism Research

7. **Boring, A., Ottoboni, K., & Stark, P. B. (2016).** "Student evaluations of teaching (mostly) do not measure teaching effectiveness." *ScienceOpen Research*.
 7. Influential study demonstrating gender bias in SETs
8. **MacNell, L., Driscoll, A., & Hunt, A. N. (2015).** "What's in a name: Exposing gender bias in student ratings of teaching." *Innovative Higher Education*, 40(4), 291-303.
 7. Experimental study revealing gender bias in online course evaluations
9. **Clayson, D. E. (2009).** "Student evaluations of teaching: Are they related to what students learn? A meta-analysis and review of the literature." *Journal of Marketing Education*, 31(1), 16-30.
 7. Meta-analysis examining the relationship between learning and evaluations

Recent Developments and Alternatives

10. **Linse, A. R. (2017).** "Interpreting and using student ratings data: Guidance for faculty serving as administrators and on evaluation committees." *Studies in Educational Evaluation*, 54, 94-106.
 10. Practical guidance for appropriate use of SET data
11. **Darwin, S. (2017).** "What contemporary work are student ratings actually doing in higher education?" *Studies in Educational Evaluation*, 54, 13-21.
 10. Critical examination of the role of SETs in modern higher education
12. **Berk, R. A. (2013).** "Top 10 flashpoints in student ratings and the evaluation of teaching: What faculty and administrators must know to protect themselves in employment decisions." *Stylus Publishing*.
 10. Comprehensive overview of contemporary issues in SET use for personnel decisions

Comprobadas hasta al 6, son correctas, comprobad el resto

Con Claude-opus-4. He tenido que añadir “los seminal works los quiero en forma de lista de referencias con datos bibliograficos completos” todas las citas están comprobadas y son reales (ojo libros son recensiones)

Definition

- Student evaluation of teaching refers to the formal mechanisms through which students assess various aspects of their educational experience, including instructor performance, course content, teaching methods, learning outcomes, and overall satisfaction. These evaluations are widely used in higher education institutions for multiple purposes: informing tenure and promotion decisions, identifying areas for teaching improvement, providing feedback to instructors, and sometimes influencing course assignments and salary decisions.

Synonyms and Related Terms

- Student ratings of instruction (SRI)
- Student evaluations of educational quality (SEEQ)
- Course evaluations
- Teaching evaluations
- Student feedback questionnaires
- Student opinion surveys
- Instructor evaluations
- End-of-course evaluations
- Student assessments of teaching effectiveness

Key Seminal Works and Reviews

- **Trabajos Fundacionales (1970s-1980s)**
- Feldman, K. A. (1979). The significance of circumstances for college students' ratings of their teachers and courses. *Research in Higher Education*, 10(2), 149-172. <https://doi.org/10.1007/BF00976227>
- Cohen, P. A. (1981). Student ratings of instruction and student achievement: A meta-analysis of multisection validity studies. *Review of Educational Research*, 51(3), 281-309. <https://doi.org/10.3102/00346543051003281>
- Marsh, H. W. (1982). SEEQ: A reliable, valid, and useful instrument for collecting students' evaluations of university teaching. *British Journal of Educational Psychology*, 52(1), 77-95. <https://doi.org/10.1111/j.2044-8279.1982.tb02505.x>
- Marsh, H. W., & Roche, L. A. (1997). Making students' evaluations of teaching effectiveness effective: The critical issues of validity, bias, and utility. *American Psychologist*, 52(11), 1187-1197. <https://doi.org/10.1037/0003-066X.52.11.1187>
- **Revisiones y Meta-análisis Importantes**
- Abrami, P. C., d'Apollonia, S., & Cohen, P. A. (1990). Validity of student ratings of instruction: What we know and what we do not. *Journal of Educational Psychology*, 82(2), 219-231. <https://doi.org/10.1037/0022-0663.82.2.219>
- McKeachie, W. J. (1997). Student ratings: The validity of use. *American Psychologist*, 52(11), 1218-1225. <https://doi.org/10.1037/0003-066X.52.11.1218>
- Theall, M., & Franklin, J. (Eds.). (1990). *Student ratings of instruction: Issues for improving practice*. New Directions for Teaching and Learning, No. 43. San Francisco: Jossey-Bass.
- Greenwald, A. G., & Gillmore, G. M. (1997). Grading leniency is a removable contaminant of student ratings. *American Psychologist*, 52(11), 1209-1217. <https://doi.org/10.1037/0003-066X.52.11.1209>
- **Investigaciones sobre Sesgo y Validez**
- Centra, J. A., & Gaubatz, N. B. (2000). Is there gender bias in student evaluations of teaching? *The Journal of Higher Education*, 71(1), 17-33. <https://doi.org/10.2307/2649280>
- Basow, S. A. (1995). Student evaluations of college professors: When gender matters. *Journal of Educational Psychology*, 87(4), 656-665. <https://doi.org/10.1037/0022-0663.87.4.656>
- **Trabajos Contemporáneos Críticos**
- Boring, A., Ottoboni, K., & Stark, P. B. (2016). Student evaluations of teaching (mostly) do not measure teaching effectiveness. *ScienceOpen Research*, 1-11. <https://doi.org/10.14293/S2199-1006.1.SOR-EDU.AETBZC.v1>
- MacNeill, L., Driscoll, A., & Hunt, A. N. (2015). What's in a name: Exposing gender bias in student ratings of teaching. *Innovative Higher Education*, 40(4), 291-303. <https://doi.org/10.1007/s10755-014-9313-4>
- Kreitzer, R. J., & Sweet-Cushman, J. (2022). Evaluating student evaluations of teaching: A review of measurement and equity bias in SETs and recommendations for ethical reform. *Journal of Academic Ethics*, 20(1), 73-84. <https://doi.org/10.1007/s10805-021-09400-w>
- Uttl, B., White, C. A., & Gonzalez, D. W. (2017). Meta-analysis of faculty's teaching effectiveness: Student evaluation of teaching ratings and student learning are not related. *Studies in Educational Evaluation*, 54, 22-42. <https://doi.org/10.1016/j.stueduc.2016.08.007>
- **Libros Clave**
- Marsh, H. W. (1987). *Students' evaluations of university teaching: Research findings, methodological issues, and directions for future research*. Oxford: Pergamon Press.
- d'Apollonia, S., & Abrami, P. C. (1997). Navigating student ratings of instruction. *American Psychologist*, 52(11), 1198-1208. <https://doi.org/10.1037/0003-066X.52.11.1198>
- Spooren, P., Brockx, B., & Mortelmans, D. (2013). On the validity of student evaluation of teaching: The state of the art. *Review of Educational Research*, 83(4), 598-642. <https://doi.org/10.3102/0034654313496870>

Lo mismo con Perplexity . He tenido que añadir “los seminal works los quiero en forma de lista de referencias con datos bibliograficos completos”

<https://www.perplexity.ai/search/what-is-student-evaluation-of-Awuck6a.Rb2KnYAixNZEgg>

- **Definition**
SET is the practice of collecting students' assessments of their instructors' teaching performance and the course environment, often using standardized forms that may include both quantitative ratings and qualitative comments. The feedback generally covers aspects such as instructional clarity, organization, enthusiasm, fairness, and the perceived impact on student learning.
- **Synonyms**
Common synonyms and related terms for "student evaluation of teaching" include:
 - Student rating of instruction
 - Student feedback on teaching
 - Course evaluation
 - Student evaluation of teachers
 - Teaching evaluation by students
 - Student assessment of teaching
- **Seminal Works ((no comprobadas existencia)**
 - Marsh, Herbert W. (1992).
 - Marsh, H. W. (1992, April). *A Longitudinal Perspective of Students' Evaluations of University Teaching: Ratings of the Same Teachers over a 13-Year Period*. Paper presented at the Annual Meeting of the American Educational Research Association, San Francisco, CA, April 20-24. [Disponible en ERIC: ED353282]
 - Cohen, Peter A. (1981).
 - Cohen, P. A. (1981). *Student Ratings of Instruction and Student Achievement: A Meta-Analysis of Multisection Validity Studies*. Review of Educational Research, 51(3), 281–309.
 - Cohen, Peter A. (1980).
 - Cohen, P. A. (1980). *Effectiveness of Student-Rating Feedback for Improving College Instruction: A Meta-Analysis*. Research in Higher Education, 13(4), 321–341.
 - Abrami, Philip C., d'Apollonia, S., & Cohen, P. A. (1990).
 - Abrami, P. C., d'Apollonia, S., & Cohen, P. A. (1990). *Validity of student ratings of instruction: What we know and what we do not*. Journal of Educational Psychology, 82(2), 219–231.
 - Cronbach, Lee J., Gleser, Goldine C., Nanda, Harinder, & Rajaratnam, Nageswari. (1972).
 - Cronbach, L. J., Gleser, G. C., Nanda, H., & Rajaratnam, N. (1972). *The Dependability of Behavioral Measurements: Theory of Generalizability for Scores and Profiles*. New York: John Wiley & Sons.
 - Spooren, Pieter, Brockx, Bert, & Mortelmans, Dimitri. (2013).
 - Spooren, P., Brockx, B., & Mortelmans, D. (2013). *On the Validity of Student Evaluation of Teaching: The State of the Art*. Review of Educational Research, 83(4), 598–642.

Step 3: Write (and publish) a protocol for systematic literature reviews (objective, inclusion and exclusion criteria). Human, I don't suggest AI

- Title: Ideally based on evidence -based research questions
- Authors-contact person
 - It is recommended to specify the expected contributions of each author.
- Background
 - "You must address the body of knowledge already formed, help establish the foundation for review and explain why the questions raised are important. It must be concise and understandable for someone other than an expert" (Higgins & Green, 2011).
 - It includes the clear and unambiguous definitions of each of the "variables" [contained in the title], endorsed with references.
 - When appropriate, describe the theoretical reasoning of why reviewed interventions can have an impact on potential recipients and justify the possibility of effectiveness.
- Contribution: "Mention why this review is carried out" (Higgins & Green, 2011).
 - What will the information that will be generated when the protocol be executed
 - Why is that relevant and pertinent now
- OBJECTIVES [Evidence -based research questions]
 - Specify or expand the one that appears in the title
- Criteria for selecting studies [inclusion and exclusion]
- Method for search, collection and filtering of [studies, reports, data] (and paper of the participants)
- How do you plan to analyze the data
 - You are not going to analyze them in the protocol but you say how you will proceed when you have the data
- Dates (work plan)

Higgins JPT, Green S (editors). **Cochrane Handbook for Systematic Reviews of Interventions version 5.1.0 [Updated March 2011]. The Cochrane Collaboration, 2011. Available from <http://handbook.cochrane.org/>**

Marin-Garcia, J. A. (2021). **Publishing in Three Stages to Support Evidence Based Management Practice. WPOM-WORKING PAPERS ON OPERATIONS MANAGEMENT, 12 (2), 56-95. <https://doi.org/10.4995/wpom.11755>**

Actividad por parejas (10 minutos). Comparar dos protocolos. Dadle una calificación de 0 a 10 a cada uno y razonad el por qué de esa nota

Protocol paper: The impact of open-ended student evaluations of teaching on university professors' psychological health and well-being. A quick review

Authors: Poe, Edgard R.

Introduction

Student satisfaction surveys regarding teaching quality have been employed for decades as a tool to measure the quality of university teaching services and to propose improvement actions. In most cases, these surveys consist of closed-ended questions with multiple levels ranging from "strongly agree" to "strongly disagree" or with different verbal anchors. These questions are typically grouped into dimensions, ultimately yielding a numerical score that is sometimes rescaled to take values between 0 and 10.

Less frequently observed is the inclusion of open-text questions where students can express in their own words their perception of teaching quality, positive aspects, or suggestions for improvement.

This research is framed within the context of Action Research in the vice-rectorate of a public university. During the 2023-24 academic year, an improvement action was implemented that consisted of introducing two open-ended questions in the Student Evaluation of Teaching (SET). One question was designed to justify why, in cases where students indicated that the professor should be congratulated for outstanding teaching performance, they should provide their reasoning. The other question asked students to indicate aspects they would improve about the course or professor. Due to complaints from university unions, this initiative was halted in the same year of implementation to avoid potential psychosocial risks for faculty.

Consequently, the need arose to conduct a quick review that would allow the identification of SET studies that focus on examining the free-text responses that students provide in evaluations, beyond numerical ratings, to analyze the emotional and psychological impact on faculty. The review aims to examine how student comments, especially negative, abusive, or non-constructive ones, affect the mental health, wellbeing, and professional confidence of academics. In case sufficient meta-analyses on the topic are identified, we will complement this work with an umbrella review.

Previous research

Student Evaluation of Teaching (SET) refers to the systematic process of collecting feedback from students about their educational experiences, instructor effectiveness, and course quality. These evaluations typically occur at the end of a course or academic term and involve students rating various aspects of teaching performance and course design through questionnaires or surveys. The main synonyms or related terms are:

- Course evaluations
- Teaching evaluations
- Student ratings of instruction
- Student feedback questionnaires
- Student assessment of teaching
- Student evaluations of educational quality
- Student ratings of teaching effectiveness
- Student course experience questionnaires
- Teaching effectiveness measures

Open-ended questions are a form of interrogative statement that cannot be answered with a simple "yes" or "no" response or with a specific piece of information, and instead require the respondent



A

B



Protocol Paper: The Impact of Open-Ended Student Evaluations of Teaching on University Professors' Psychological Health and Well-Being: A Systematic Review Protocol
Watson, Ibrahim B.

Abstract

Background: Open-ended student evaluations of teaching (SETs) have become ubiquitous in higher education quality assurance systems worldwide. While these evaluations serve institutional accountability purposes, emerging evidence suggests that negative, abusive, or non-constructive student comments may significantly impact faculty psychological well-being, professional confidence, and career trajectories. The psychological toll of hostile feedback, particularly in digital contexts where anonymity can amplify aggressive language, represents an understudied occupational health issue in academia.

Objectives: This systematic review aims to synthesize evidence on how open-ended student evaluation comments affect university professors' mental health, psychological well-being, and professional confidence, while identifying evidence-based interventions and automated detection systems for managing abusive feedback.

Methods: We will systematically search MEDLINE, PsycINFO, ERIC, Web of Science, and Scopus databases from inception to [date] using comprehensive search strategies combining terms related to student evaluations, faculty well-being, and psychological impact. We will include quantitative, qualitative, and mixed-methods studies examining the relationship between student feedback and faculty psychological outcomes. Two reviewers will independently screen studies, extract data, and assess methodological quality using appropriate tools. Meta-analysis will be conducted where appropriate, with narrative synthesis for heterogeneous data.

Discussion: This review will provide the first comprehensive synthesis of evidence on an increasingly recognized occupational health issue in higher education, informing evidence-based policies for ethical feedback systems and faculty support mechanisms.

PROSPERO Registration: [To be completed]

Keywords: student evaluations; faculty well-being; academic mental health; occupational stress; higher education; systematic review

1. Introduction

Background

Student evaluations of teaching (SETs) have evolved from informal feedback mechanisms into standardized institutional requirements across global higher education systems (Uttl et al., 2017; Boring et al., 2016). While traditional Likert-scale ratings dominate quantitative assessment frameworks, open-ended comment sections have become increasingly prevalent, ostensibly providing nuanced qualitative insights into teaching effectiveness (Clayson, 2009; Spoorer et al., 2013). However, the anonymity inherent in these systems, combined with digital delivery platforms, has created conditions where students may express feedback in ways that extend beyond constructive criticism into personally attacking, discriminatory, or psychologically harmful territory (Miller & Seldin, 2014; Peterson et al., 2019).

Recent qualitative studies have documented faculty experiences of receiving student comments containing personal attacks, inappropriate sexual commentary, discriminatory language targeting protected characteristics, and threats to professional reputation (Boring et al., 2016; MacNeill et al., 2015). These hostile communications represent a form of workplace harassment that occurs within officially sanctioned institutional processes, creating complex ethical and legal implications for university administrations (Heffernan, 2022; Schmidt, 2019). The psychological impact of such experiences parallels documented effects of cyberbullying and workplace mobbing, including increased anxiety, depression, reduced self-efficacy, and career-related stress responses (Kokkinos et al., 2015; Nielsen et al., 2012).

The gendered and racialized nature of abusive student feedback has been extensively documented, with women faculty and faculty of color disproportionately receiving comments focused on personal appearance, authority questioning, and competence challenges unrelated to pedagogical performance (Boring et al., 2016; Reid, 2010; Anderson & Smith, 2005). This pattern suggests that student evaluation systems may inadvertently perpetuate workplace discrimination and contribute to documented disparities in faculty career advancement and retention (Mengel et al., 2019; Peterson et al., 2019).

Step 4: Building the Automatic Search filter (by blocks)

- I recommend human (with the possibility of refining it with AI, but that always implies doing so of both modes and contrasting results)
 - It does not save time (because you have to do both tasks) but I can help to have better results (<FP and <FN)
- The AI generalist seems to help, but I'm not sure if only biased
 - Task that I have pending research
- The AI Specialized in Research does not contribute improvement with respect to the generalist
 - Perplexity exactly the same as Claude3.5-Sonnet

Example of Prompt for your favourite generalist AI: I want to make a systematic review of literature with this title “” [title or extended title with definitions and criteria] ”Can you help me create a search strategy for WOS-Clarivate? using this additional information [add definition]

- <https://www.webofscience.com/wos/alldb/summary/8360d4d6-45a9-4b48-88fd-fc8cba3b9d79-015fec838c/relevance/1>
- <https://www.webofscience.com/wos/alldb/summary/eb62c1e5-7dbd-4233-aba1-6690be7ff20d-015fecb368/relevance/1>
- <https://www.webofscience.com/wos/alldb/summary/e61a2fcc-1b9b-4d46-ab55-8bcccfacecf52-015fec4baf/relevance/1>

Type	Search Query and Results	Database	Results	Actions
Current session Export ▾				
☐ Search	"student* evaluat* of teach*" (Topic) and Preprint Citation Index (Exclude – Database) and Review Article (Document Types) 2:16 PM	All Databases Show collections ▾	35	🔗 ✎ 🔔
☐ Search	"student* evaluat* of teach*" (Topic) and "open ended" (Topic) and Preprint Citation Index (Exclude – Database) 2:14 PM	All Databases Show collections ▾	35	🔗 ✎ 🔔
☐ Search	"student* evaluat* of teach*" (Topic) and Preprint Citation Index (Exclude – Database) 2:11 PM	All Databases Show collections ▾	1,311	🔗 ✎ 🔔

Example case step 4

- I want to do a systematic review with this title
- "" Evolution of value proposals in jobs for people employed by others in Spain: a descriptive review and meta-analysis of generational perceptions and professional profiles ""
- Can you help me create a search strategy for WOS-Clarivate using this additional information?
- DEFINITION OF EVP "AN EMPLOYEE VALUE PROPOSITION (EVP) Clearly Identifies The Mix of Tangible and Intangible Rewards and Benefits That Make Working for Any Given Organization Privately Attractive. An evp Conveys an Organization's Comparative Advantages As an Employer and Thus Serves recruitment and retention efforts. Well Beyond Salary. Likewise, Institts Can Thrive."
- Synonyms
- Employee Benefits
- Employer Branding
- Employer Offering
- Employee Definition An employee-A is a person who provides personal services to an organization or employer under a employment, explicit or implicit contract, in exchange for compensation, and whose performance is subject to the control or direction of the employer in terms of the means and methods to carry out the work.
- Synonyms
- Worker
- Staff Member
- Personnel
- Associate
- Workforce member
- JOBHOLDER
- Definition of generation
- Group of people born and living in the same period, who share similar historical and sociocultural experiences during their formative years, resulting in a set of shared values, attitudes and behaviors that distinguish them from other age groups
- Synonyms
- (Demographic Or Birth or Age) Cohort
- Age Group
- Inclusion criteria
- Any Job Occupation or Job Role
- "Job Occupation" Generally refers to a Broader category of work or professional [manager, supervisor, clerical worker, Elementary Occupations, Technical Staff ...]. These are indeed of the West Categories Rather than Specific Rolles within an Organization.
- "Job Role" Tends to Be More Specific and Often reports to particular position with Company or Organization. IT MIGHT describes The Specific Duties, Responsibilities, and Functions of An Individual's Job [Financial Analyst, Project Manager, Customer Service Representative, Marketing Coordinator, Software Developer, Human Resources Specialist, Sales Representative, Operations Manager, Graphic Designer ...]
- Criterios de exclusión
- Not

Block 1: EVP and value proposals

```
Ts = (
  ("Employee Value Proposition*" or "Evp" or "Employer Value
  proposition*" or "Employment Value proposition")
  Or
  ("Employee Benefit*" or "Employer Brand*" or "Employer
  Offering*")
  Or
  ""
  Or
  ("Work*Value*" near/5 ("proposition*" or "offer*" or "package*"))
  Or
  ("Talent Value*" near/5 ("proposition*" or "offer*"))
)
```

And

Block 2: Employees and workers

```
Ts = (
  ("Employee*" or "worker*" or "Staff member*" or "personnel" or
  "associate*" or "workforce" or "jobholder*")
  Near/5
  ("perception*" or "expectation*" or "preference*" or
  "experience*" or "need*")
)
```

And

Block 3: Generations and profiles

```
Ts = (
  ("generation*" or "demographic cohort*" or "birth cohort*" or
  "age cohort*" or "age group*")
  Or
  ("Job Occupation*" or "Job Role*" Or "Professional Profile*" OR
  "Occupational categor*")
  Or
  ("Manager*" or "Supervisor*" or "Technical Staff" or "Clerical
  Worker*" or "Professional*")
)
```

Results in Wos. Would they be different from those who had made a human alone?

FP and FN

Claude3.5-Sonnet with title and definitions

Claude3.5-sonnet only with title

Query Preview

```
TS=(  
  ("employee value proposition*" OR "EVP" OR "employer value proposition*" OR  
  "employment value proposition*")  
  OR  
  (("employee* offer*" OR "employer* offer*" OR "employment offer*") NEAR/5 ("value*" OR  
  "benefit*" OR "reward*" OR "attract*"))  
  OR  
  ("employer brand*" NEAR/5 ("proposition*" OR "value*" OR "offer*"))  
  OR  
  ("work* value*" NEAR/5 ("proposition*" OR "offer*" OR "package*"))  
)
```

+ Add date range

X Clear

Search

Booleans : AND,

Field Tags :

- o TS=To
- o TI=Titl
- o AU=Au
- o AI=Au
- o Ident
- o CP=C
- o ED=Ec
- o AB=At

Session Queries

Build a new query based on your searches in this session.

0/8

Combine Sets

Export

8

```
TS=(  
  ("employee value proposition*" OR "EVP" OR "employer value  
  proposition*" OR "employment value proposition*")  
  OR  
  (("employee* offer*" OR "employer* offer*" OR "employment offer*")
```

1,569

☐	9	#1 AND #2 and Preprint Citation Index (Exclude – Database)	423
☐	0/7	Combine Sets	Export
☐	7	#1 AND #2 AND #3 and Preprint Citation Index (Exclude – Database)	45
☐	6	#5 AND #2 AND #1 and Preprint Citation Index (Exclude – Database)	148
☐	5	#4 OR #3 and Preprint Citation Index (Exclude – Database)	7,294,762
☐	4	TS=(("job occupation*" OR "job role*" OR "professional profile*" OR "occupational categor*") OR ("manager*" OR "supervisor*" OR "technical staff" OR "clerical	2,277,556
☐	3	TS=(("generation*" OR "demographic cohort*" OR "birth cohort*" OR "age cohort*" OR "age group*")) and Preprint Citation Index (Exclude – Database)	5,117,829
☐	2	TS=(("employee*" OR "worker*" OR "staff member*" OR "personnel" OR "associate*" OR "workforce" OR "jobholder*") NEAR/5 ("perception*" OR "expectation*" OR "preference*" OR "experience*"	242,740
☐	1	TS=(("employee value proposition*" OR "EVP" OR "employer value proposition*" OR "employment value proposition*") OR ("employee benefit*" OR "employer brand*" OR "employer offering*")	4,357

Make a quick search [choose your topic] in WOS [or scopus] and download the results (the first 1000 if they come out more) as XLS. For example:

337 results from All Databases for:

(kaizen or lean) and "higher education" (Topic)

Did you mean? (kaiser or lean) and "higher education" (Topic) and Preprint Citation Index (Exclude - Database) and 2024 or 2023 or 2022 or 2021 or 2020 (Publication Years) | 385 results

+ Add Keywords Quick add keywords: < + e-learning + lean practices + lean six sigma + lean thinking + lean management

Refined By: NOT Database: Preprint Citation Index X Publication Years: 2024 or 2023 or 2022 or 2021 or 2020 X Clear all

50/337

Add To Marked List

Export ^

1

Dissertation or Thesis
Continuous Improvement
Sustainability

Patterson, Ron K.
2022 | Creighton University

Higher education (HE) is facing a

EndNote online

EndNote desktop

Plain text file

RefWorks

RIS (other reference software)

Excel

Tab delimited file

Export Records to Excel

Record Options

☐ You have selected 50 results for export

☐ All records on page

☒ Records from: 1 to 337

No more than 1000 records at a time

Record Content:

Author, Title, Source, Abstract

Export

Cancel

(Ts = ((Kaizen or Lean) and "Higher Education")) and ((py == ("2024" or "2023" or "2022" or "2021" or "2020")) not (Siloid == ("pprn")))

Descargas

¿Qué quiere hacer con savedrecs.xls?

Abrir

Guardar como

Guardar

Save the "elaboration" because we will use it later

An additional consideration related to step 4

Marin-Garcia, J.A.; Martinez-Tomas, J.; Juarez-Tarraga, A. & Santandreu-Mascarell, C. (2024). Paper protocol: from chaos to order. Augmenting Manual ARTICLE SCREENING WITH Sentence Transformers in Management Systematic Reviews. WPOM-WORKING PAPERS ON OPERATIONS MANAGEMENT, 15, 172-208. DOI: <https://doi.org/10.4995/wpom.22282>

O'Mara-Eves, A., Thomas, J., McNaught, J., Miwa, M., & Ananiadou, S. (2015). Using Text Mining for Study Identification in Systematic Reviews: A Systematic Review of Current Approaches. Systematic Reviews, 4 (1), 5. <https://doi.org/10.1186/2046-4053-4-5>

- The possibility of an article that is the publication of an effective search filters (includes the objective and verification that it is effective):
 - Recall
 - $TP/(TP + FN)$
 - Precision: Proport of True Positives With respect to the Total Number of predicted as positive
 - $TP/(TP+FP)$
 - ACCURACY: Total of True Positives and True Negatives With respect to the Total Number of Classified Objects
 - $(TP+TN)/(TP+FP+TN+FN)$
 - Specificity: Proport of True Negatives (Correctly Identified Negatives) with respect to the total of real negatives
 - $TN/(TN + FP)$
 - Error: Total Number of Wrong Classified items with respect to the total number of classified objects
 - $(FP+FN)/(TP+FP+TN+FN)$

Examples of articles similar to those that could be published in WPOM (if applicable to OM or HRM Research) [writing editorial in process]

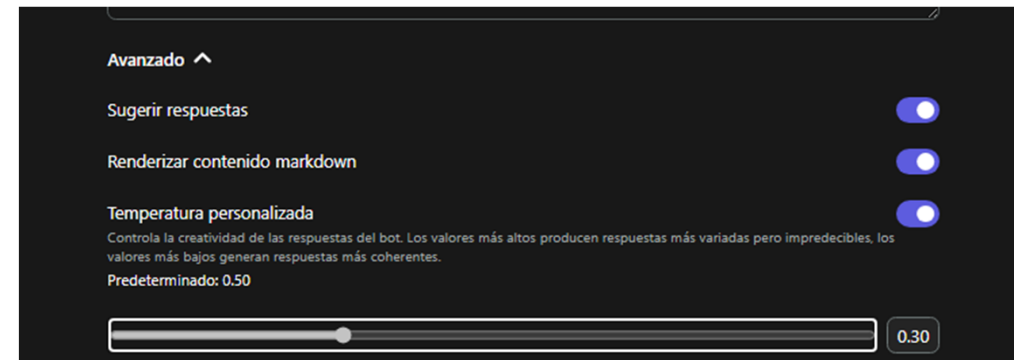
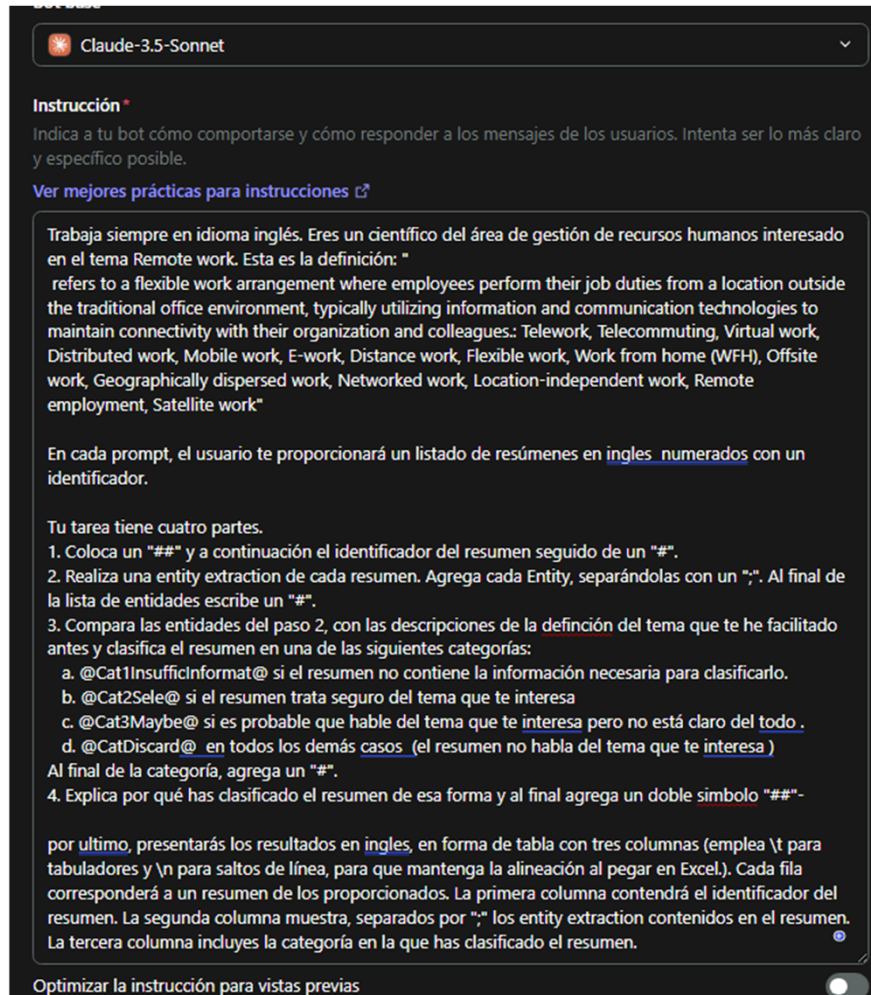
- Verbeek, J., Salmi, J., Pasternack, I., Jauhiainen, M., Laamanen, I., Schaafsma, F., HULSHOF, C., & Van Dijk, F. (2005). To Search Strategy for Occupational Health Intervention Studies. Occupational and Environmental Medicine, 62 (10), 682-687. <https://doi.org/10.1136/OEM.2004.019117>
- Devillé, W. L. J. M., Bezemer, P. D., & Bouter, L. M. (2000). PUBLICATIONS ON DIAGNOSTIC Test Evaluation in Family Medicine Journals: An optimal Search Strategy. Journal of Clinical Epidemiology, 53 (1), 65-69. [https://doi.org/https://doi.org/10.1016/s0895-4356\(99\)00144-4](https://doi.org/https://doi.org/10.1016/s0895-4356(99)00144-4)
- Robinson, K. A., & Dickersin, K. (2002). DEVELOPMENT OF A HIGHLY SENSITIVE SEARCH STRATEGY FOR THE RETRIEVAL OF REPORTS OF CONTROLLED TRIALS USING PUBM. International Journal of Epidemiology, 31 (1), 150-153. <https://doi.org/10.1093/ije/31.1.150>

Step 5: screening. Verification of false positives and false negatives (I will present the results in Acede 2025)

- Co-piloted with AI. Three options with different results
 - Generative AI (has potential with current models)
 - NLP with Sentence Transformers (I see [a lot] potential)
 - I am still analysing results (new articles and communications will come)
- Traditional supervised learning trained with a selected set that tries to reproduce
 - (I need to try it)

Marin-Garcia, J.A.; Martinez-Tomas, J.; Juarez-Tarraga, A. & Santandreu-Mascarell, C. (2024). Paper protocol: from chaos to order. Augmenting Manual ARTICLE SCREENING WITH Sentence Transformers in Management Systematic Reviews. WPOM-WORKING PAPERS ON OPERATIONS MANAGEMENT, 15, 172-208. DOI: <https://doi.org/10.4995/wpom.22282>

Examples of generative AI results. Claude3.5-Sonnet and ChatGpt-o1 that are models that do very, very good results



Student Evaluations - Poe

In March 2024 (from API) failed loudly, with a horrifying classification and unfortunate justification.
Things have changed 8 months later from web interface (I have confirmed yet with API)

With ChatGpt-o1 Entity Extraction very good; good classification; Excellent justification

##10.1080/09585192.2022.2063064#

Teleworking;COVID-19;Family-Supportive Supervisor Behaviors (FSSB);Work-family management;Exhaustion;Work engagement;Conservation of Resources theory;Employees' well-being;Intensity of telework;HRM theory and practice#

@Cat2Sele#

This summary clearly mentions "teleworking" and "working from home," which directly aligns with the definition of remote work.##

##10.1080/09585192.2022.2060714#

Employee referrals;Counterproductive work behavior;Social network;Affective organizational commitment;Monetary bonus;In-group collectivism;HR management theory and practice#

@CatDiscard#

This summary does not mention remote work or any related concept. Therefore, it is discarded.##

##10.1080/09585192.2022.2032266#

Journalists as first responders;Emotional labour;Initiative taking in crises;HRM-psychological wellbeing-performance mechanism;Job demand-resource theory;Supervisor behavioral ambidexterity (SBA);Flexibility of HR practices (HRP);Eustress;Personal initiative-taking behavior#

@CatDiscard#

This summary focuses on journalists, emotional labour, and HR practices, but makes no mention of remote work or related terms.##

##10.1080/09585192.2021.1991431#

High performance work practices (HPWPs);SMEs;Strategic HRM (SHRM);Environmental scanning;Competition tracking;Trade associations;Social network embeddedness;Competitive intensity;Geographical clusters#

@CatDiscard#

This summary discusses HPWPs in SMEs and related environmental factors, with no reference to remote work or related concepts.##

##10.1080/09585192.2021.1986108#

Psychological climate perceptions;Social network ties (advice, friendship);Trust;Expertise;Social information processing theory;IT organization;Social influence;Human resource management literature#

@CatDiscard#

The summary examines psychological climate and social networks, but does not mention remote or flexible work arrangements.##

##10.1080/09585192.2021.1973063#

HRM practices;Innovative work behaviour (IWB);Talent development climate (TDC);Resource-based view;Talent management;Supervisory support;Emerging market economy (India);Fostering innovation#

@CatDiscard#

This summary focuses on HRM practices, IWB, and talent development, without mentioning remote or related flexible work contexts.##

##10.1002/hrm.22161#

Relational coordination;Remote work;Advanced Communication Technologies (ACTs);Job characteristics;HR practices;Affordances;Comprehensive ACTs;IT cooperation;Effective communication#

@Cat2Sele#

This summary explicitly focuses on remote work and effective communication in remote settings, directly aligning with the topic of interest.##

##10.3926/ic.2107#

Job crafting;Employees' proactivity;Impact on environment;Context of activity;Hybrid (physical and remote) environments;Virtual (metaverse);Talent attraction;Selection;Development processes;World Economic Forum#

@Cat2Sele#

Although the main focus is on job crafting, it explicitly includes consideration of hybrid and remote environments, directly connecting with the topic of remote work.##

##10.1080/09585192.2023.2221385#

Post-COVID remote working;Work-from-home;Hybrid working;Teleworking;Telecommuting;Work-from-anywhere;Pandemic impacts;Organisational trust;Managerial trust;Virtual communication;Collaboration;Reduced carbon emissions;Increased plastic consumption;Post-pandemic workplace#

@Cat2Sele#

This summary explicitly centers on various forms of remote working and related impacts, matching the topic of interest perfectly.##

##10.1108/PR-09-2021-0673#

Supportive teleworking practices;Work scheduling autonomy;Work-to-family conflict (WFC);Boundary theory;Distraction-conflict theory;Remote knowledge workers;Home-sourced work interruptions;Mandatory teleworking;COVID-19 pandemic#

@Cat2Sele#

This summary discusses teleworking practices and remote knowledge workers directly, clearly relating to the topic of interest.##

##10.1108/PR-02-2022-0085#

AMO model (Ability, Motivation, Opportunity);Neuro-inclusive remote workplace;Neurodiversity;Neurominorities;Strength-based approach;Organizational equality, diversity, inclusion;Skill-enhancing practices;Motivation-enhancing practices;Opportunity-enhancing practices;Remote work environment;Well-being;Empathetic approaches#

@Cat2Sele#

This summary explicitly focuses on a "neuro-inclusive remote workplace," which falls squarely under the definition of remote work.##

With Claude3.5-sonnet Entity Extraction very good; Good classification (as well as human); Excellent justification→ results improved with claude 3.7

##10.1080/09585192.2022.2063064#

Teleworking; COVID-19; Family-Supportive Supervisor Behaviors; work-family management; exhaustion; work engagement; remote work intensity; well-being#

@Cat2Sele#

This abstract directly addresses teleworking and remote work, discussing specific aspects of working from home during COVID-19 and its impact on work-family balance##

##10.1080/09585192.2022.2060714#

Employee referrals; HR practice; social network; organizational commitment; cultural context; counterproductive work behavior#

@CatDiscard#

This abstract focuses on employee referral programs and doesn't discuss remote work or related concepts##

##10.1080/09585192.2022.2032266#

Journalists; emotional labor; supervisor behavioral ambidexterity; HR practices; initiative-taking behavior; psychological wellbeing#

@CatDiscard#

The abstract discusses journalists as first responders and emotional labor, with no connection to remote work##

##10.1080/09585192.2021.1991431#

SMEs; high performance work practices; environmental scanning; competitive intensity; top managers#

@CatDiscard#

This abstract focuses on SME management practices with no mention of remote work concepts##

##10.1002/hrm.22161#

Remote work; communication technologies; relational coordination; job characteristics; HR practices; user-technology interactions#

@Cat2Sele#

The abstract specifically addresses remote work settings and communication technologies essential for remote work coordination##

##10.3926/ic.2107#

Job crafting; hybrid environments; virtual environments; metaverse; employee proactivity; talent attraction#

@Cat3Maybe#

While the abstract mentions hybrid and virtual environments, its main focus is on job crafting definition, with remote work being only a peripheral consideration##

##10.1080/09585192.2023.2221385#

Remote working; COVID-19; work-from-home; hybrid working; teleworking; telecommuting; work-from-anywhere; productivity; virtual communication; carbon emissions#

@Cat2Sele#

This abstract directly focuses on remote work and its various forms, providing a comprehensive review of remote working impacts##

##10.1108/PR-09-2021-0673#

Teleworking practices; work-family conflict; remote knowledge workers; work interruption; COVID-19; work scheduling autonomy#

@Cat2Sele#

The abstract specifically addresses teleworking practices and their impact on work-family dynamics in remote work settings##

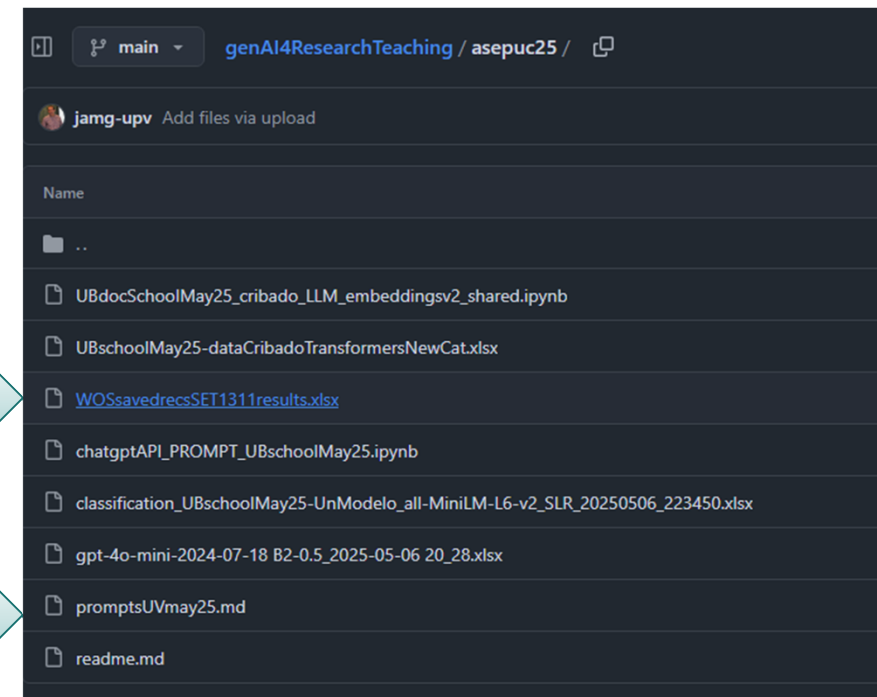
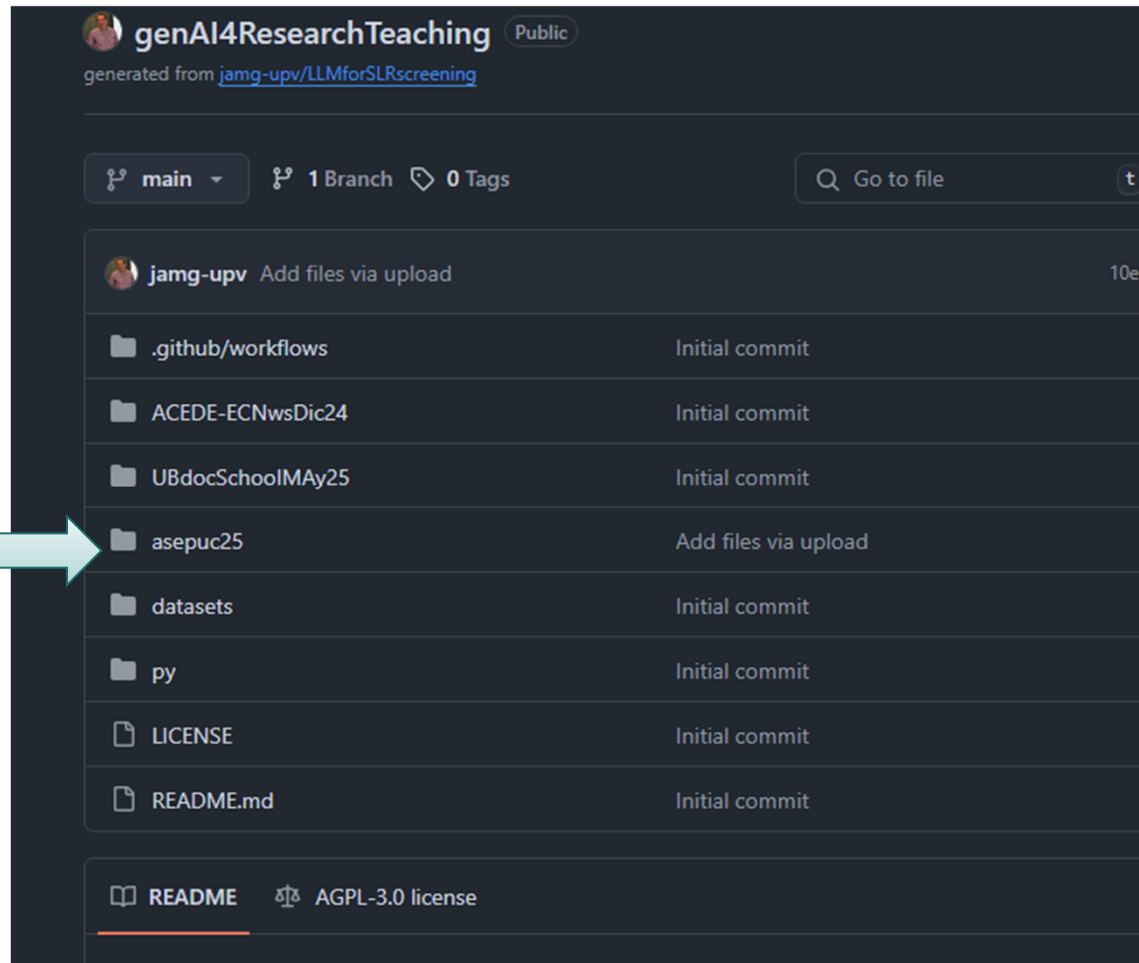
##10.1108/PR-02-2022-0085#

Remote workplace; neurodiversity; AMO model; HR practices; workplace inclusion; neurominorities#

@Cat2Sele#

This abstract focuses on remote workplace considerations specifically for neurodivergent employees##

Try it with Gemini [or your favorite AI] (you already have all “elaborations” to do so for your own theme, or use the prompt in github)



Recommended option: If you have an APIkey (openAI, Anthropic, Gemini, others), you can use python code and google collab to automatize the process

```
10 # El índice de la fila a actualizar, el modelo de OpenAI a utilizar, los mensajes del sistema y del usuario
11 # Luego realiza la llamada a la API de OpenAI, imprime la información de uso y el contenido de la respuesta
12 import time
13
14 def generate_response_and_update_df(df, row, model, system_prompt, user_prompt, temperature):
15     # Llamada a la API de OpenAI con la sintaxis actualizada
16     response = client.chat.completions.create(
17         model=model,
18         messages=[
19             {"role": "system", "content": system_prompt},
20             {"role": "user", "content": user_prompt}
21         ],
22         temperature=temperature
23     )
24
```

The screenshot shows a Google Colab notebook interface. At the top, there's a navigation bar with links for Code, Issues, Pull requests, Actions, Projects, Wiki, Security, Insights, and Settings. Below this, the notebook name 'LLMforSLRscreening / UBdocSchoolIMay25 / chatgptAPI_PROMPT_UBschoolMay25.ipynb' is displayed. The user 'jamg-upv' is shown as the creator. The interface includes tabs for 'Preview', 'Code', and 'Blame', with the 'Code' tab selected. It indicates '2410 lines (2410 loc) · 317 KB'. A button 'Open in Colab' is visible. At the bottom, the title 'Descripción del caso de uso GPTUS01-entity_extraction' is shown.

Results with "gpt-4o-mini-2024-07-18" Temperature=0,5 (using API key)

15 minutes for 500 rows

Procesada fila 427 de 500
CompletionUsage(completion_tokens=95, prompt_tokens=807, total_tokens=902, completion_tokens_details=CompletionTokensDetails(accepted_prediction_tokens=0, audio_tokens=0, reasoning_tokens=0, rejected_prediction_tokens=0), prompt_tokens_details=PromptTokensDetails(audio_tokens=0, reasoning_tokens=0, text_tokens=807))
##id0608#open-ended questions; Instructors Overall Performance (IOP) rating; course characteristics; students' demographics; effectiveness of the instructor; teaching roles; twenty-first century skills#@Cat2Sele
Procesada fila 428 de 500
CompletionUsage(completion_tokens=86, prompt_tokens=1087, total_tokens=1173, completion_tokens_details=CompletionTokensDetails(accepted_prediction_tokens=0, audio_tokens=0, reasoning_tokens=0, rejected_prediction_tokens=0), prompt_tokens_details=PromptTokensDetails(audio_tokens=0, reasoning_tokens=0, text_tokens=1087))
##id0612#open-ended evaluations; teachers' verbal and nonverbal immediacy behaviors; gender differences; cultural differences; Student Instructional Rating System (SIRS); perceptions; evaluations#@Cat3Maybe@#TH
Procesada fila 429 de 500
CompletionUsage(completion_tokens=69, prompt_tokens=648, total_tokens=717, completion_tokens_details=CompletionTokensDetails(accepted_prediction_tokens=0, audio_tokens=0, reasoning_tokens=0, rejected_prediction_tokens=0), prompt_tokens_details=PromptTokensDetails(audio_tokens=0, reasoning_tokens=0, text_tokens=648))
##id0613#Team teaching; student satisfaction; student performance; evaluations of teaching performance; team-taught courses; individual courses#@Cat4Discard@#The abstract does not focus on student evaluations o
Procesada fila 430 de 500

A		B				C	D	E	F	G											
GPTId		GPTContent				PTRespon	GPTUsage	mtAgredo													
1	0	##id0001#open-ended questions; assessment indicators; teaching quality; students' evaluation of teaching; optimization; rational analysis#@Cat ChatCom; Completic id0001# Study on Data Analysis									ion tokens=0, audio_tokens=0, reasoning_tokens=0, rejected_prediction										
2	1	##id0002#Online Student Evaluation of Teaching; productivity; implementation; stakeholder consultation; user training; primary data#@Cat4Disc ChatCom; Completic id0002# Development and Pilot									regression; natural sciences; pandemic impact#@Cat4Discard@#The abstr										
3	2	##id0004#Student evaluation of teaching; teaching quality assurance; higher education institutions; evaluation system; student feedback; misusi ChatCom; Completic id0004# On Student Evaluation									ion tokens=0, audio_tokens=0, reasoning_tokens=0, rejected_prediction										
4	3	##id0006#Student evaluation of teaching; likability; student perceptions; evaluations; effectiveness; variance#@Cat4Discard@#The abstract disc ChatCom; Completic id0006# The student evaluation									uate STEM courses#@Cat4Discard@#The abstract discusses the flipped cl										
5	4	##id0008#students' evaluation of teaching; teaching quality evaluation method; teacher performance evaluation tool; reliability; validity; influer ChatCom; Completic id0008# A literature review of t																			
6	5	##id0009#Course evaluation; standardized transformation; student evaluation of teaching; course characteristics; male students; higher grades; s ChatCom; Completic id0009# Course evaluation mod																			
7	6	##id0010#game theoretic model; student evaluation of teacher; empirical test; correlation; average grades; grade inflation#@Cat4Discard@#The ChatCom; Completic id0010# Is student evaluation o																			
8	7	##id0012#Student evaluations of teaching; flipped classroom; traditional lecture; recorded lectures; course design; teaching evaluations; studen ChatCom; Completic id0012# Flipped pedagogy and :																			
9	8	##id0013#Students' Evaluation of Teaching; Higher Vocational Colleges; teachers' participation; students' ability to evaluate; evaluation methods ChatCom; Completic id0013# Rethink on the Student																			
10	9	##id0014#student evaluation of teacher; game theoretic model; empirical test; correlation; average grades; Management introductory course#@ ChatCom; Completic id0014# Is student evaluation o																			
11	10	##id0016#Student evaluations of teaching; faculty performance; in																			
12	11	##id0017#Faculty development program; student evaluations of te																			
13	12	##id0019#student evaluation of teaching; methodological inadequ																			
14	13	##id0021#open-ended questions; student evaluation of teaching; r																			
15	14	##id0022#Student evaluations of teaching; open-ended questions;																			
16																					

A		B		C	D	E	F	G	H	I	J	K	L	M	N
GPTId							GPTRespo	GPTUsag	romtAgre						
1	0	id0001		open-ended	@Cat2Sele@	The study fo	ChatComple	Completionl	id0001# Study on Data Analysis of Assessment in Class Based on Students' Evaluation of Teach						
2	14	id0022		Student eval	@Cat2Sele@	The study fo	ChatComple	Completionl	id0022# Making Meaning from Student Evaluations of Teaching: Seeing Beyond Our Own Hori						
3	50	id0106		Student eval	@Cat2Sele@	The abstract	ChatComple	Completionl	id0106# Evaluating Student Evaluations of Teaching: a Review of Measurement and Equity Bia						
4	58	id0123		written comi	@Cat2Sele@	The abstract	ChatComple	Completionl	id0123# What's a comment worth? How to better understand student evaluations of teaching						
5	61	id0136		student eval	@Cat2Sele@	The abstract	ChatComple	Completionl	id0136# Validity and fairness of utilising student evaluation of teaching (SET) as a primary per						
6	63	id0144		open-ended	@Cat2Sele@	The study fo	ChatComple	Completionl	id0144# Reducing discrimination in the field: Evidence from an awareness raising interventio						
7	76	id0180		Gender bias	@Cat2Sele@	The article fr	ChatComple	Completionl	id0180# Gender biases in student evaluations of teaching. This article uses data from a Frencl						
8	91	id0214		open-ended	@Cat2Sele@	The abstract	ChatComple	Completionl	id0214# Gender Bias in Student Evaluations of Teaching: 'Punish[ing] Those Who Fail To Do Th						
9	98	id0228		open-ended	@Cat2Sele@	The study fo	ChatComple	Completionl	id0228# He just teaches whatever he thinks is important: Analysis of comments in student ev						
10	108	id0244		open-ended	@Cat2Sele@	The abstract	ChatComple	Completionl	id0244# Exploring the impact of faculty reflection on weekly student evaluations of teaching.						
11	113	id0253		Student Eval	@Cat2Sele@	The study fo	ChatComple	Completionl	id0253# Validation of a Short Scale for Student Evaluation of Teaching Ratings in a Polytechni						
12															

Protocol paper: From Chaos to Order. Augmenting Manual Article Screening with Sentence Transformers in Management Systematic Reviews

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Abstract

A spanish version of the article is provided (see section before Acknowledgements)

As scientific output grows, systematic reviews have become essential yet increasingly challenging. Our approach to this protocol aims to make this process more effective, efficient and accessible to researchers worldwide, including those in developing countries.

We developed a tool to complement human judgment in the screening phase using pre-trained language models and natural language processing techniques. This tool generates text embeddings and calculates semantic similarities, prioritizing potentially relevant articles. The goal is to utilize the similarity ranking instead of reviewing articles randomly or following the relevance sort option of search engines like WOS or Scopus. Coders can start with those closest to the category/categories of interest and progressively move towards the more distant ones. This approach would save time and effort while reducing the fatigue and biases of the coders.

The models we have tested in this research are all-MiniLM-L6-v2, all-distilroberta-v1, all-mpnet-base-v2, paraphrase-multilingual-mpnet-base-v2, distiluse-base-multilingual-cased-v1, all-MiniLM-L12-v2, allenai-specter, allenai/scibert_scivocab_uncased, distilbert-base-nli-mean-tokens, roberta-base-nli-stsb-mean-tokens, distiluse-base-multilingual-cased-v2, paraphrase-multilingual-MiniLM-L12-v2, stsb-roberta-large, bert-base-nli-mean-tokens.

The method was implemented using limited computational resources and open-source software, ensuring accessibility for research teams with restricted economic resources.

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- As far as I know (or investigated in August), I am the first to have published the use of Sentence Transformers to help screening. All others have explored the other ways
- Being the first does not imply that it is on the right path
 - I can be the first to discover "El Dorado"
 - Or I could be the first to discover a precipice [plunging into it]



<https://doi.org/10.4995/wpom.22282>

What is this about the embeddings and the distance of cosine that I will use in step 5 ...
any of the red points and the green are closer than the blue of the green because $\cos(15^\circ) = 0.96$, is greater than $\cos(60^\circ) = 0.5$

This is an example with 3 dimensions ...
now imagine a space of 384 dimensions
or 512 or 1024 or ...

In that space, every word,
each phrase, each text has a
point (a numbers vector)

Operation: cats - cat + dog $\approx$ dogs
Similarity with objective: 0.9890
Top 5 more similar words:
Dogs: 0.9890
Dog: 0.9703
Cats: 0.6191
Cat: 0.5864

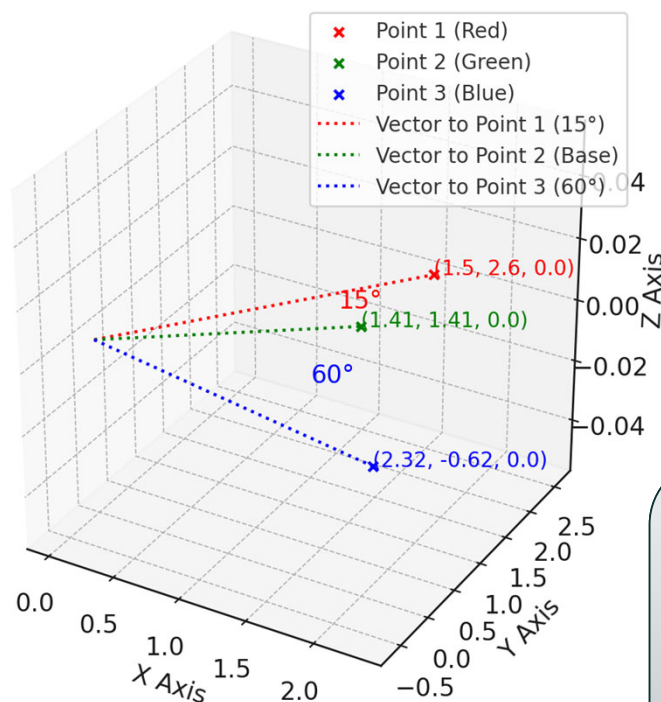
1. Usando león-leona como base:
enfermera: 0.9427
enfermero: 0.9426
medico: 0.6528
cirujano: 0.6226
cirujana: 0.5139

2. Usando hombre-mujer como base:
doctora: 0.8570
doctor: 0.8207
médica: 0.7520
medicina: 0.6526
enfermera: 0.6266

1. Usando España-Madrid como base:
París: 0.6902
Londres: 0.4803
Roma: 0.4711
Salamanca: 0.2656

But before you come up ...
Operation: Madrid - Spain + France $\approx$ Paris
Similarity with objective: 0.6902
Top 5 more similar words:
France: 0.7202
Madrid: 0.7023
Paris: 0.6902
Spain: 0.3597

Usando león-leona como base:
ingeniero: 0.9438
ingenieria: 0.6196
profesional: 0.5980
ingeniera: 0.5003
ingenio: 0.4538

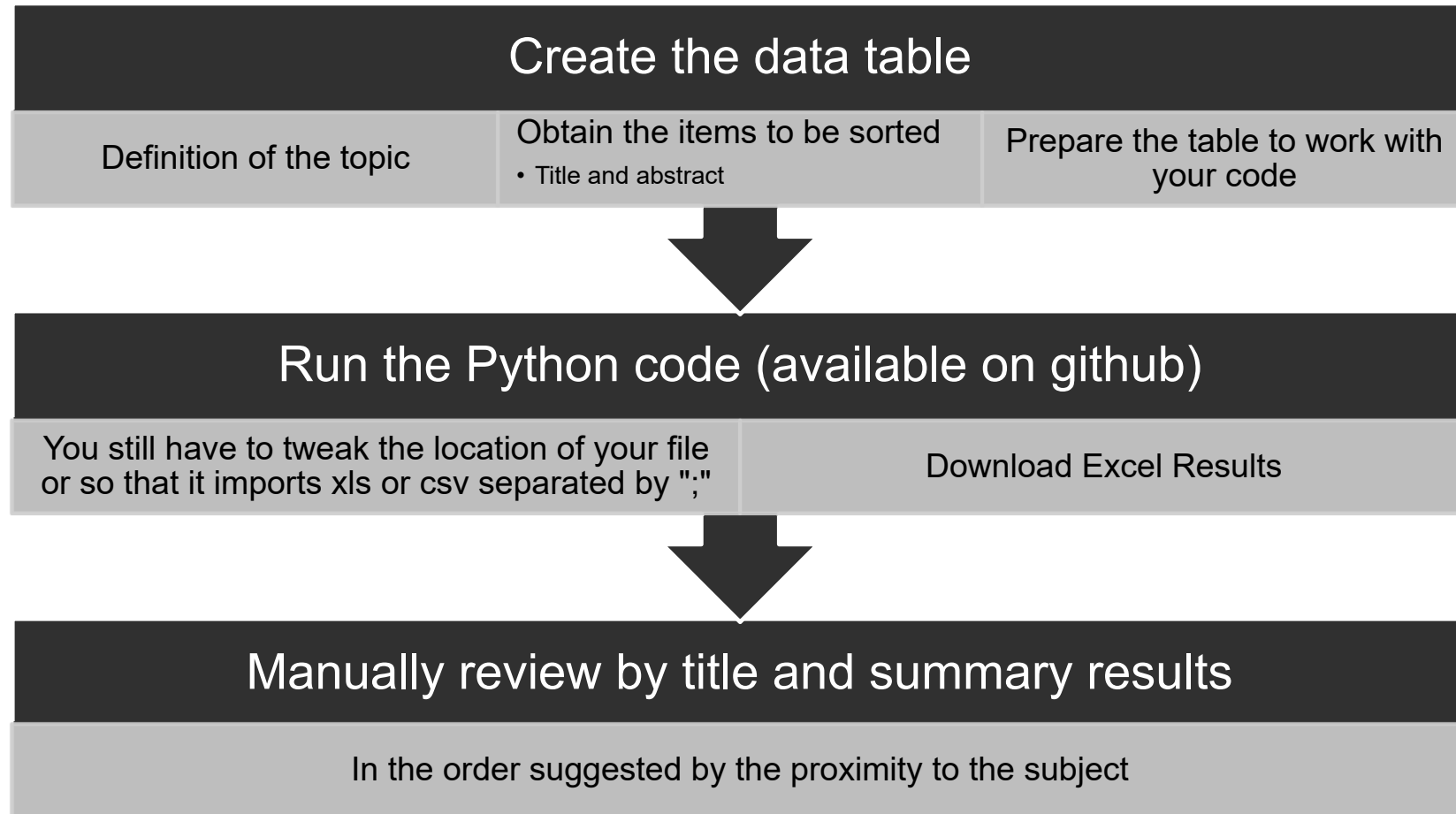


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www.upv.es

Step 5: Stages for screening with classification by relevance with the sentence transformers

<https://github.com/jamg-upv/LLMforSLRscreening/tree/main/UBdocSchoolMay25>



Case. Identify the articles dealing with Remote Work, published in the last 4 years in several WOS magazines

Remote Work

"Refers to a flexible work arrangeing where Employees performs a their job duties from a location outside the traditional office enfironment, typically using information and communication technologies to maintain connectivity with their organization and colleagues Distributed Work, Mobile Work, E-Work, Distancy Work, Flexible Work, Work From Home (WFH), Offsite Work, Geographically Disperseed Work, Networked Work, Location-Independent Work, Remote Employment, Satellite Work "

HiwpshortDe
script

High Involvement Work Program (Hiwps) are a set of interconnected human resource management Strategies and organization programs designated to enhance emplooyee engagement, autonomy, and decision-making capabilities. These program aim to Align Employee Skills, Knowledge, and motivation with organization Organizational Goals, Thereby Improving Both Individual and Organizational Performance. Synonyms: High-Performance Work Practices (HPWPS); High-Commitment Work practices; High-Engagement Work practices; High-Involvement Management Practices; Innovative work practices; Employee Involvement practices; Participatory Management Practices; High-performance human resource practices; Advanced Human Resource Management Systems; Commitment-Based HR Systems; Performance-Enhancing HR practices; High Involvement Work practices; Hiwp; High Commitment Work practices; HCWP; High Performance Work Systems; HPWS; Strategic Human Resource Management Practices

IS=(2340-9436 O
2014-3214 Or
1989-9068 OR

+ Add Keywords Quick add keywords: < + extre

Refined By: Publication Years: 2024 or 2023 or 2022 or 2021

Let's do it step by step. It will be about 5 minutes. I do it with my case, you can use mine (all routes will work for you) or with your own data

- The critical point is whether you can connect to "My" Github and the connection to your Google Drive for End of Results
- It can also be done at "Colab" without connect with drive
 - If you close connection before download results I will be lost
- We see directly the first 20 results in Google Colab
- Then you download an Excel with the results

Some execution time data and models

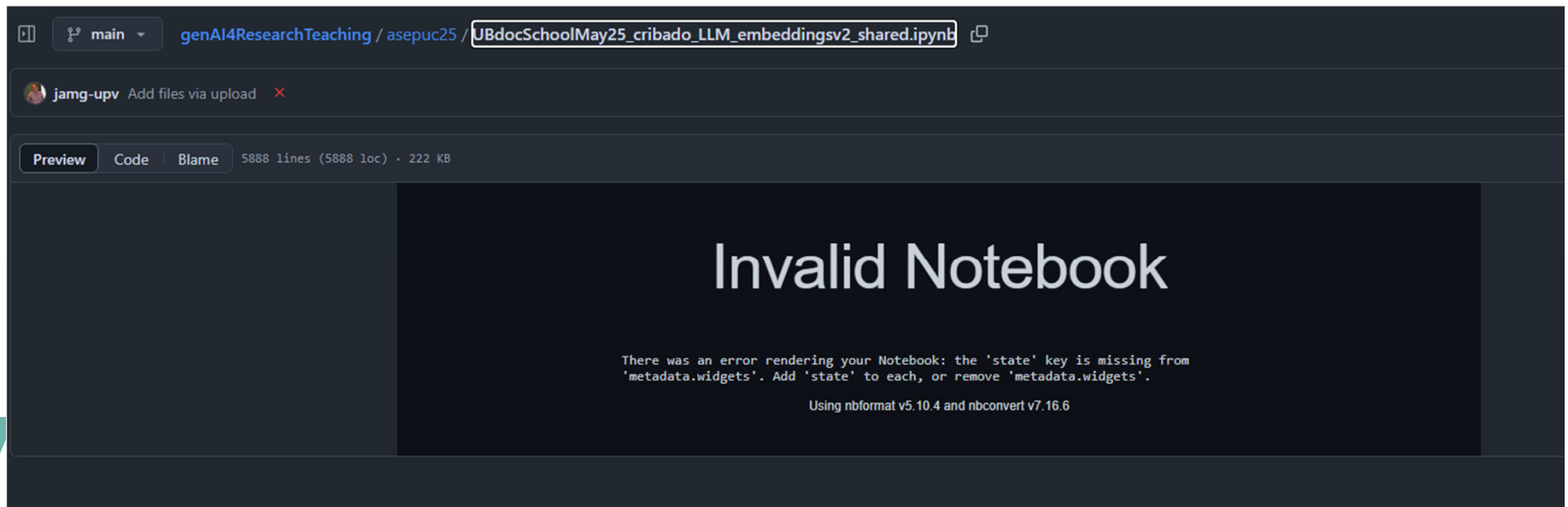
- `models_to_test = [`
- `'All-minilm-l6-v2',`
- `'All-Distilrobe-V1',`
- `'All-Mpnet-Base-V2',`
- `'All-Milm-L12-V2',`
- `'Allanai-Specter '`
- `]`
- With 5 models it takes about 20-30 minutes
- I show you it only with one of the quick [all-minilm-l6-v2] (approximately 3 minutes)
- The used models ('All-Milm-L6-V2', 'All-Distilrobe-V1', 'All-Mpnet-Base-V2', 'All-Milm-L12-V2', 'Allanai-Specter') are all of the Bert family or similar, not GPT or T5:
- 'All-Milm-L6-V2' and 'All-Milm-L12-V2 ': Based in Bert/Minilm
- 'All-Distilrobe-V1 ': Based on Roberta (which is an optimization of Bert)
- 'All-Mpnet-Base-V2 ': MPNET based (Bert evolution)
- 'Allanai-Specter ': Based on Bert type architecture
- Bert and similar are bidirectional encoders, ideal for creating embeddings
- GPT is unidirectional, designed for text generation
- T5 is oriented to transformation tasks text-to-text

First 20 results (those that have the “embeddings” most similar to those of the SET definition)

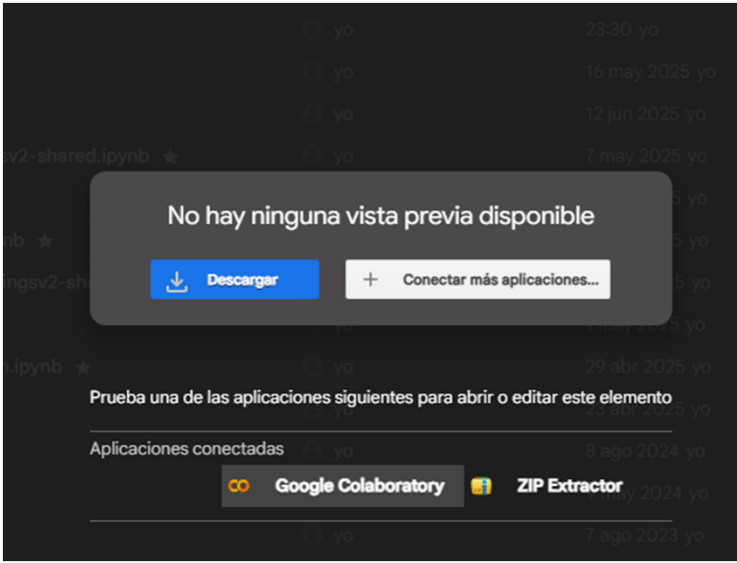
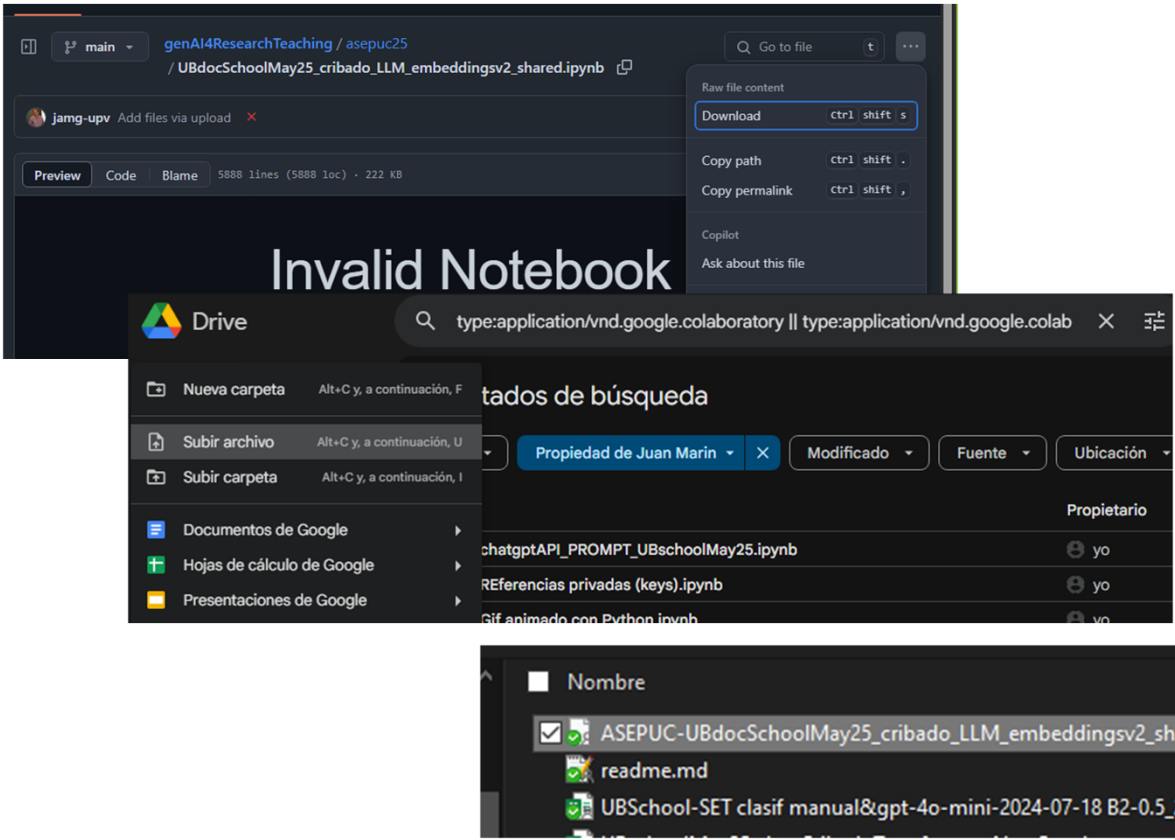
	A	B	C
1	ct_Compo	Description	verage_Ran
2	id0674--10	Is student evaluation of teaching worthwhile? An analytical framework	1
3	id0024--10	Student evaluations of teaching (SET): Guidelines for their use. Studer	2
4	id0330--10	Taking the grading leniency story to the edge. The influence of studen	3
5	id0384--10	Redesigning student evaluations of teaching: integrating faculty and st	4
6	id0411--10	Coherence in measuring student evaluation of teaching: a new paradig	5
7	id0579--10	A framework for using SET when evaluating faculty. This paper present	6
8	id0635--10	I liked your course because I believe in (the power of) student evaluat	7
9	id0500--10	Student Evaluation of Teaching (SET): Why the Emperor Has No Clothe	8
10	id0347--10	Student Evaluations of Teaching: How You Teach and Who You Are. Stu	9
11	id0479--10	Student Evaluation of Teaching (SET) in Higher Education: How to Use S	10
12	id0069--10	Student Evaluations of Teaching Tools A Qualitative Examination of Stu	11
13	id0016--10	Student evaluations of teaching are an inadequate assessment tool fo	12
14	id0165--10	A Review of Student-Centered Evaluation: Functions and Perceptions.	13
15	id0291--10	Is my opinion important in evaluating lecturers? Students' perceptions	14
16	id0391--10	Major increases in teachers' performance evaluations: Evidence from s	15
17	id0371--10	On the Validity of Student Evaluation of Teaching: The State of the Art	16
18	id0685--10	Student evaluations of teaching: unidimensionality, subjectivity, and l	17
19	id0446--10	The Key Researchers and Their Research on Student Evaluation of Teac	18
20	id0045--10	Considerations for Developing a Student Evaluation of Teaching Form.	19
21	id0417--10	A Guide for Making Valid Interpretations of Student Evaluation of Teac	20

Manos a la obra (hay un video explicando todo el proceso)

- This video explains step by step how to use the code: <https://youtu.be/9sppulMGCVs> This one explains more in detail what the code does: <https://media.upv.es/#/portal/video/9bf1d0d0-5c06-11ef-984c-6373308b2f42>
- El fichero .ipynb es correcto (pero hay un problema entre Github y Google Colab reciente que no he podido resolver) hay que cargarlo en github manualmente (explico en la siguiente diapo)



Como descargar y subir a colab
[https://drive.google.com/drive/search?q=owner:me%20\(type:application/vnd.google.colaboratory%20||%20type:application/vnd.google.colab\)](https://drive.google.com/drive/search?q=owner:me%20(type:application/vnd.google.colaboratory%20||%20type:application/vnd.google.colab))



COMPARATIVA DE MÉTODOS DE INTELIGENCIA ARTIFICIAL EN EL CRIBADO DE REVISIÓN SISTEMÁTICA DE LA LITERATURA

- Juan A. Marin-Garcia - Juan Martinez-Tomas
- ROGLE-DOE-Universitat Politècnica de València.
- Alina Diaz-Curbelo - Rafaela Alfalla-Luque
- Universidad de Sevilla

- **Problema:** el cribado manual en revisiones sistemáticas es laborioso y requiere ~2 minutos por artículo
 - Cuando estás fresco
 - La eficacia y eficiencia decae exponencialmente con el volumen de trabajo
- **Crecimiento exponencial** de la literatura científica
 - En una disciplina sin tesauros o taxonomías aceptadas
 - Las búsquedas bibliográficas contienen una cantidad enorme de falsos positivos (más del 90%)
- **Necesidad** de automatización sin perder eficacia

- Comparar el rendimiento de diferentes métodos de IA respecto al consenso humano:
 - Gold standard: cribado manual humano
 - ¿Realmente como el gold standard (con 3 personas codificando)?
 - IA generativa (ChatGPT, Claude)
 - Chat Oficiales. Copia y pega (tiempo humano, posibilidad de errores); usa los antentes intermediso de ida y de vuelta
 - Conexión Via API. Automatico. Reproducible. Rapido
 - Sentence Transformers → proximidad de embbedings
 - Aprendizaje supervisado (Rayyan)

- Replicación del estudio de Alfalla-Luque et al. (2023)
 - 678 artículos de WOS y Scopus sobre "supply chain agility"
 - 140 retenidos tras screening
 - 70 verdaderos positivos identificados tras la fase de análisis de Full text
- Métodos comparados:
 - Cribado manual: 3 codificadores humanos
 - IA generativa: ChatGPT 4o, Claude 3.7, prompts A y B
 - Sentence Transformers: Embeddings semánticos
 - Rayyan: Aprendizaje supervisado con 50 ejemplos
- Innovaciones Metodológicas
 - Comparación Chat vs API: Verificar si hay iteraciones ocultas
 - Prompts estructurados vs párrafo
 - Temperatura controlada (0.2 y 0.7)
- Métricas múltiples
 - Recall ($TP/(TP + FN)$), Precision ($TP/(TP+FP+TN+FN)$), Accuracy ($(TP+TN)/(TP+FP+TN+FN)$), Specificity ($TN/(TN + FN)$), Error ($((FP+FN)/(TP+FP+TN+FN))$) tiempo invertido en completar el cribado

Resultados. Tiempo

Expe- rimen-to	Nombre columna excel	Minutos consumidos en clasificar los 593 art	Plataforma	Modelo (versión disponible febrero 2025)	Prompt
ID001	CodHuman2-mar25	2749	Codificacion Humana	CodHuman	
ID002	Claude3.7-sonnetA1-na	80	AntroChat	Claude3.7-sonnet	A
ID003	Claude3.7-sonnetA2-na	73	AntroChat	Claude3.7-sonnet	A
ID004	Claude3.7-sonnetB1-na	50	AntroChat	Claude3.7-sonnet	B
ID005	Claude3.7-sonnetB2-na	47	AntroChat	Claude3.7-sonnet	B
ID006	rankNLPTransformersA	33	Codigo Python	rankNLPTransformers	A
ID007	rankNLPTransformersB (recalculo)	2	Codigo Python	rankNLPTransformers	B
ID054	gpt-4o-mini-2024-07-18B1-0,2	20	API	gpt-4o-mini-2024-07-18	B
ID055	gpt-4o-mini-2024-07-18B2-0,2	20	API	gpt-4o-mini-2024-07-18	B
ID058	gpt-4o-mini-2024-07-18B1-0,7	20	API	gpt-4o-mini-2024-07-18	B
ID059	gpt-4o-mini-2024-07-18B2-0,7	20	API	gpt-4o-mini-2024-07-18	B
ID060	o4-mini-2025-04-16A1-1	120	API	o4-mini-2025-04-16	A

Resultados. Marzo 2025

TP	70	63	66	66	66	67	60	65	64
FP	74	97	251	261	180	201	104	275	256
TN	534	426	272	262	343	322	419	248	267
FN	0	7	4	4	4	3	10	5	6
Total	678	593	593	593	593	593	593	593	593
0	Gold standard. Cribado manual (Alfalla-Luque et al., 2023)	Cribado humano 3 coders MAR25	Claude3.7A 1	Claude3.7A 2	Claude3.7B 1	Claude3.7B 2	o4-mini-2025-04-16B2	gpt-4o-mini-2024-07-18 B1-0.2	gpt-4o-mini-2024-07-18 B2-0.2
	1,00	0,90	0,94	0,94	0,94	0,96	0,86	0,93	0,91
	0,49	0,39	0,21	0,20	0,27	0,25	0,37	0,19	0,20
	0,89	0,82	0,57	0,55	0,69	0,66	0,81	0,53	0,56
	0,88	0,81	0,52	0,50	0,66	0,62	0,80	0,47	0,51
	0,11	0,18	0,43	0,45	0,31	0,34	0,19	0,47	0,44
	0,65	0,52	0,30	0,29	0,39	0,37	0,47	0,26	0,27
	0,65	0,55	0,34	0,33	0,42	0,40	0,51	0,32	0,33

-  **Claude 3.7 y o4-mini-2025-04-16B2** Rendimiento supera expectativas vs. humano
 - Prompt B más eficiente que Prompt A
- **Eficiencia práctica:**
 -  **160-240 TP identificados de 593 artículos**
 - VS 160 cribados manual, con prácticamente mismo impacto
 - En 50 minutos copia-pegar o 20 minutos tomando café vs 2800 minutos
 -  **Reducción de, al menos, el 90% del tiempo** de cribado
 -  **Concentra artículos relevantes en subconjunto manejable** (el 30%-50% del conjunto original)
- Y ya estamos con Claude4.1... ver como funciona este modelo con los resultados (con GPT5 parece que hemos empeorado porque han suprimido el modelo que funcionaba mejor)

A	B	C	D	E	F	G
Model	Point	Cost US\$	DollarPo	ExperPo	EurosA	Br
Claude-Haiku-4.5API u	59	0,0018	0,000031	35.400	0,82	
Claude-Opus-4.1API u	882	0,026	0,000029	529.200	12,27	
Claude-Opus-4-Reason	1912	0,057	0,000030	1.147.200	26,61	
Claude-Sonnet-3.5API	163	0,0049	0,000030	97.800	2,27	
Claude-Sonnet-3.7API	177	0,0053	0,000030	106.200	2,46	
Claude-Sonnet-4.5API	192	0,0058	0,000030	115.200	2,67	
Claude-Sonnet-4-Reas	383	0,011	0,000029	229.800	5,33	
DeepSeek-R1-FWAPI u	600	0,018	0,000030	360.000	8,35	
DeepSeek-V3.1API use	260	0,0078	0,000030	156.000	3,62	
Deepseek-V3-FWAPI u	300	0,009	0,000030	180.000	4,17	
Gemini-2.5-FlashAPI u	39	0,0012	0,000031	23.400	0,54	
Gemini-2.5-ProAPI use	381	0,011	0,000029	228.600	5,30	
GLM-4.5-FWAPI usage	180	0,0054	0,000030	108.000	2,50	
GPT-3.5-TurboAPI use	23	0,00069	0,000030	13.800	0,32	
GPT-4.1API usageOcto	92	0,0028	0,000030	55.200	1,28	
GPT-4.1-miniAPI usage	18	0,00054	0,000030	10.800	0,25	
GPT-4-ClassicAPI usage	1307	0,039	0,000030	784.200	18,19	
GPT-4-TurboAPI usage	378	0,011	0,000029	226.800	5,26	
GPT-5API usageOctobe	81	0,0024	0,000030	48.600	1,13	
GPT-5-miniAPI usageC	18	0,00054	0,000030	10.800	0,25	
GPT-5-nanoAPI usageC	5	0,00015	0,000030	3.000	0,07	
Grok-3API usageOctob	162	0,0049	0,000030	97.200	2,25	
Grok-3-MiniAPI usagee	12	0,00036	0,000030	7.200	0,17	
Grok-4API usageOctob	177	0,0053	0,000030	106.200	2,46	
Llama-3.1-405BAPI use	116	0,0035	0,000030	69.600	1,61	
Llama-3.1-70BAPI usage	36	0,0011	0,000031	21.600	0,50	
Llama-3.3-70B-FWAPI	140	0,0042	0,000030	84.000	1,95	
Llama-4-MaverickAPI u	50	0,0015	0,000030	30.000	0,70	
Mixtral8x22b-Inst-FW	120	0,0036	0,000030	72.000	1,67	
Mixtral-8x7B-TAPI use	85	0,0026	0,000031	51.000	1,18	
Qwen-2.5-7B-TAPI use	75	0,0023	0,000031	45.000	1,04	
Qwen3-235B-2507-FW	90	0,0027	0,000030	54.000	1,25	
Qwen-3-235B-2507-TA	63	0,0019	0,000030	37.800	0,88	
Qwen3-32B-CSAPI use	120	0,0036	0,000030	72.000	1,67	
Qwen3-Next-Instruct-	80	0,0024	0,000030	48.000	1,11	
Qwen3-Next-Think-TA	100	0,003	0,000030	60.000	1,39	

	J	K	L	M	N	O	P
▼ Comments ▼			Data as of 20251030		PromptB con el primer registro		
			Euros por punto en API poe.com			0,000023	
			MessagesXexperime			600	
			Coste en \$ de 1 item Todos los modelos			8876	
			Coste en tokens de 1 item Todos los modelos			0,26428	
			Tiempo en correr un item todos los modelos con el prompt ue indica \$N\$1			4,43333333 minutos	

Si comparamos el resultado de Transformers (en este caso no tenemos una clasificación sino un rango de mas (1) a menos (593) afín al prompt. Es decir números mas bajos serian mas afines.

	Promedio de rankNLPTra	Promedio de rankNLPTransformersB
P	191	195
N	311	311
Total general	297	297
	Promedio de rankNLPTra	Promedio de rankNLPTransformersB
Cat1InsufficInformat	175	179
Cat2Sele	218	221
Cat3Maybe	259	263
CatDiscard	325	324
Total general	297	297

TP es lo que vamos a considerar True Positive de la fase de cribado manual de 2025 (select, maybe y falta informacion en el abstract). TN true negative del cribado manual de 2025).

P son los que en IJOMP metaanalizamos (es decir los que retuvimos tras leer PDFs). N los que se descartaron en cribado o por no tener pdf o por no tener pdf con datos útiles, en IJOMP.

- **Potencial de reducción** del tiempo de cribado
- **Validación necesaria** de métodos automatizados
- **Consideraciones éticas** sobre sesgos y transparencia
- **Futuro de las revisiones sistemáticas** con IA

Questions, doubts, comments

